OSIPI · GSOC 2026 · ASL QUALITY CONTROL

Renal ASL Explained Simply

A from-zero guide to arterial spin labelling in the kidney, written for someone who already knows brain ASL

Agnik Misra

KIDNEY_ASL_EXPLAINED.md · 17,992 words

reconciled against osipy-qc · 20 registered checks · 253 tests

🫘 Renal ASL — Explained Simply

A from-zero guide to arterial spin labelling in the kidney, written for someone who already knows brain ASL

> 🧭 **What this file is:** the kidney equivalent of `../RESEARCH_EXPLAINED.md` and `../WHITEPAPER_EXPLAINED.md`. It assumes you know what a PLD is and why M0 matters, and spends its time on **everything the kidney does differently.** By the end you should be able to sit in a meeting with a renal perfusion specialist and not embarrass yourself.
>
> 📌 **Companion files:** `papers/README.md` (the 51-PDF source library + the citation errors I found), `papers/` (the PDFs themselves), `screenshots/` (68 rendered pages at 130 dpi).
>
> 🔒 **The rule this document obeys:** every number carries its **ROI, cohort, field strength and labelling scheme.** A renal perfusion value without its ROI is meaningless — cortex and medulla differ by a factor of ~2.5–7 depending on how you draw the mask. Where I could not verify something, it says so.

📋 Part 0 — The one-page version

If you read nothing else:

1. **The kidney has its own White Paper.** It is **Nery et al. 2020, MAGMA 33(1):141–161** — a modified-Delphi consensus from a **23-member expert panel** (26 authors) that produced **59 agreed statements** and 2 that failed. It is CC-BY and sits at `papers/nery2020_renal_asl_consensus.pdf`. **David Alsop co-authored both it and the 2015 brain White Paper** — the lineage is direct.

2. **Both FAIR and PCASL are endorsed.** The brain toolbox's "assume PCASL" default is invalid here. Statement **R3.1** (6% abstention, 93% agreement): "*Both PASL:FAIR and PCASL are adequate labeling strategies.*"

3. **The default readout is 2D SINGLE-SLICE spin-echo EPI, coronal oblique.** Not 3D. Statement **R6.3** explicitly says 3D is *not* recommended as the default. This inverts the brain White Paper.

4. **Free breathing, not breath-hold. R7.3** (0%, 94%): breath-hold scans are *not* recommended. Respiratory motion — not head motion — is the dominant failure mode.

5. **Report CORTICAL perfusion, left and right kidney separately** (**R10.1**, 0% abstention, **100% agreement** — the strongest statement in the document). And **R10.2** (14%, 89%): "*Medullary renal blood flow values are not considered reliable with current measurement approaches.*"

6. **The M0 rule is identical to brain.** M0 mandatory (**R9.1**, 94%), acquired **without labelling and without background suppression**, long TR, corrected if TR is not > 5 s. Your existing Module 6 ports across almost unchanged.

7. **The consensus contains ZERO numeric image-quality thresholds.** No tSNR floor, no CoV cutoff, no motion limit, no QEI analogue. I searched the full text: `mL/100` appears exactly once, in the phrase "*typically mL/100 g/min*", with no value attached. **Every renal Stream-B threshold you ship starts life**

`uncalibrated`. That is not a weakness of the design — it is the honest state of the field, and it is the gap the toolbox exists to fill.

PART 1 — What renal ASL is, and why the kidney is harder

1.1 The one-paragraph version (assuming you know brain ASL)

Everything you know still applies: tag the water in arterial blood with an RF pulse, wait for it to arrive, image, image again without tagging, subtract, divide by M0, multiply by constants. What changes is **where you tag** (the abdominal aorta, not the carotids — or, for FAIR, you invert everything *except* a slab around the kidney), **what you wait for** (the kidney is very close to the aorta, so transit is short), **what moves** (the organ itself, several centimetres, on every breath), and **what you divide the image into afterwards** (cortex and medulla, with no probability maps to help you).

The output is called **RBF — renal blood flow** — in the same units you already use, **mL/100 g/min**. The OSIPI living ASL Lexicon defines it that way explicitly: "*Kidney | Renal Blood Flow / Renal perfusion | RBF | mL/100g/min | Tissue perfusion value in kidney*".

1.2 The five steps, and what changes at each

STEP	BRAIN	KIDNEY	WHY IT MATTERS
1. LABEL	PCASL plane in the neck	**Either** PCASL plane on the **aorta**, 8–10 cm superior to the kidney centre, perpendicular to the vessel (**R5.3**, **R5.4**) — **or** FAIR, where a selective inversion slab covering the kidney must *exclude* the aorta (**R4.2**, 100% agreement)	Two completely different geometry checks, chosen by labelling scheme
2. WAIT	PLD ~1.8 s (adults)	**PCASL PLD 1.2–1.5 s** (**R5.11**, 100% agreement); **FAIR TI 1.8–2.0 s** with bolus TI1 1.0–1.2 s (**R4.4**, **R4.6**)	Shorter than brain, because the kidney sits right off the aorta
3. ACQUIRE	3D GRASE / stack-of-spirals	**2D single-slice spin-echo EPI**, coronal oblique (**R6.1**, **R6.4**, **R6.7**)	The brain's 3D default is explicitly *not* recommended for kidney
4. CONTROL	same, alternating	same, alternating, **≥20 pairs** (**R4.7** / **R5.12**)	Note the 20-pair rule is scoped in the body text to *"when using the recommended 2D readout"*
5. SUBTRACT + CALIBRATE	÷ M0, single-compartment	÷ M0, single-compartment (**R9.2**, 100% agreement), λ = 0.9 mL/g, $T1_{blood}$ 1.65 s @3T	Same maths; two constants differ (see §4.3)

1.3 Three things that make the kidney genuinely harder

(a) The organ moves, and it moves a lot

The consensus puts it bluntly:

> *"Subject breathing induces kidney displacements up to an order of magnitude larger than the typical ASL voxel size, which if unaccounted for cause a significant loss of image quality in ASL."* — Nery 2020

That is rhetoric, so here is the measurement. **Yamashita et al. 2014** tracked the **centre of gravity** of 39 kidneys in 20 cancer patients (median age 47, range 26–86) on **4D-CT** — not MRI, so treat it as displacement physics rather than an ASL result:

AXIS	DISPLACEMENT	RANGE
Cranio-caudal	**11.1 ± 4.8 mm**	2.5 – 20.5 mm
Anterior-posterior	3.6 ± 2.1 mm	0.6 – 8.0 mm
Right-left	1.7 ± 1.4 mm	0.4 – 5.9 mm

Motion correlated with respiratory phase at **$r > 0.97$, $p < 0.01$ in all three directions**.

Yamashita 2014 renal motion by axis

A second cohort disagrees in magnitude, and the reason is instructive. Siva et al. 2013 (62 analysed of 71 patients, median age 68) measured the **kidney apex**, not the centre of gravity, and got mean cranio-caudal displacement of **0.74 cm left / 0.75 cm right** (ranges 0.10–2.15 and 0.11–1.92 cm), with 90th percentiles of 1.33 and 1.30 cm. Different landmark, different number — which is exactly the kind of thing that must be recorded alongside a value.

> **Two honest caveats before you use these.** (1) The consensus's "order of magnitude larger than the voxel" is not consistent with its own recommended in-plane resolution of **2–4 mm** (**R6.10**): 11.1 mm is about **3–5 voxels**, not 10–20. Lead with the measured number, not the rhetoric. (2) Siva 2013 also *cites* larger MRI figures (Bussels: 1.61 cm right / 1.69 cm left, free breathing, n=12) and forced-breathing extremes (Moerland: 6.6 cm left, 8.6 cm right). I have read those only as citations inside Siva, not in the originals — treat them as secondary.

The motion is also **anisotropic** by roughly 6.5 : 2 : 1 (CC : AP : RL). An isotropic scalar like brain framewise displacement would average away the one axis that actually matters.

(b) There is no GM/WM/CSF probability map for the kidney

This is the single biggest structural break, and the reason **QEI does not port**.

In the brain you get, free with any T1w scan, three continuous probability maps. QEI's structural term is literally `Pearson(CBF, 2.5·GM + 1.0·WM)` — it needs those maps.

In the kidney:

- The consensus's default is **manual ROI drawing** (**R9.10**, 0% abstention, **100% agreement**), on the **M0 image or a separately acquired structural dataset** (**R9.11**, 93%).
- The contrast that actually separates cortex from medulla is a **quantitative T1 map** — and the consensus does *not* put a T1 map in the minimum protocol.
- Every off-the-shelf renal segmentation tool I checked (UKAT, Renal Segmentor, TotalSegmentator) outputs **whole kidney only**. Renal Segmentor *can* emit a per-voxel probability map, but it is a whole-kidney probability, not a cortex/medulla one.
- The best published automatic cortex segmentation (Bones et al. 2022, 3-U-net cascade, cortex extracted from the **T1 map**) scores **Dice 0.78 ± 0.04** against reference — while a **second human expert** scores **0.77 ± 0.02** against the same reference. The cortex/medulla boundary is intrinsically ill-defined.

Bones 2022 automatic renal cortex segmentation performance

> ⛔ **Do not convert that 0.78 Dice into "22% error on the perfusion value."** Dice is a surface- overlap statistic. In the same paper, a Dice of 0.78 produced a cortical RBF difference of only **1.4%** (auto 211 ± 31 vs manual 208 ± 31 mL/min/100 g, n=10 healthy, 1.5T balanced pCASL) — statistically significant at P = .032, but small.

(c) The labelling scheme changes the answer by ~80%

Harteveld et al. 2020 scanned the **same subjects** with **both** FAIR and pCASL in the same session (16 recruited, 15 analysed at visit 1; 8 male; age **51 ± 10 y**, with a **>40 years inclusion criterion**; eGFR 86 ± 15; 3T Philips Ingenia; multi-delay; **single-shot gradient-echo EPI 2D multislice**):

	FAIR	PCASL
Cortex RBF (mL/min/100 g)	**362 ± 57**	**201 ± 72**
Medulla RBF	140 ± 47	84 ± 27
Cortex ATT (s)	0.47 ± 0.13	0.71 ± 0.25
Medulla ATT (s)	0.70 ± 0.10	0.86 ± 0.12
Cortex within-subject CV	**9.9%** (ICC 0.51)	**33.9%** (ICC −0.38)

Harteveld 2020 FAIR vs pCASL, same subjects

362 / 201 = **1.80**. That is larger than any age effect, any field-strength effect, and any disease effect in this whole document.

> 🚩 **Design consequence:** a renal perfusion plausibility band **must be keyed on labelling scheme first**, then field strength, then population. A single global band is meaningless.
>
> ⚠️ **Do not over-read the ATT difference.** The same paper says explicitly: "*The ATT is dependent on the used measurement method, and was, therefore, not directly compared between both labeling approaches.*" Part of the FAIR/pCASL ATT gap is definitional — FAIR makes the label essentially at the target, pCASL makes it 8–10 cm upstream — and part is sampling, since FAIR studies sample first delays at 0.1–0.3 s while pCASL starts at 0.4–0.5 s, and you cannot fit an ATT below your first delay.

1.4 One thing that makes the kidney *easier*

The signal is bigger. In the brain the perfusion-weighted difference is **~1%** of the control image. The Odudu 2018 review states renal cortex is "*a typical renal cortex perfusion-weighted signal intensity of 5% of the control image signal.*"

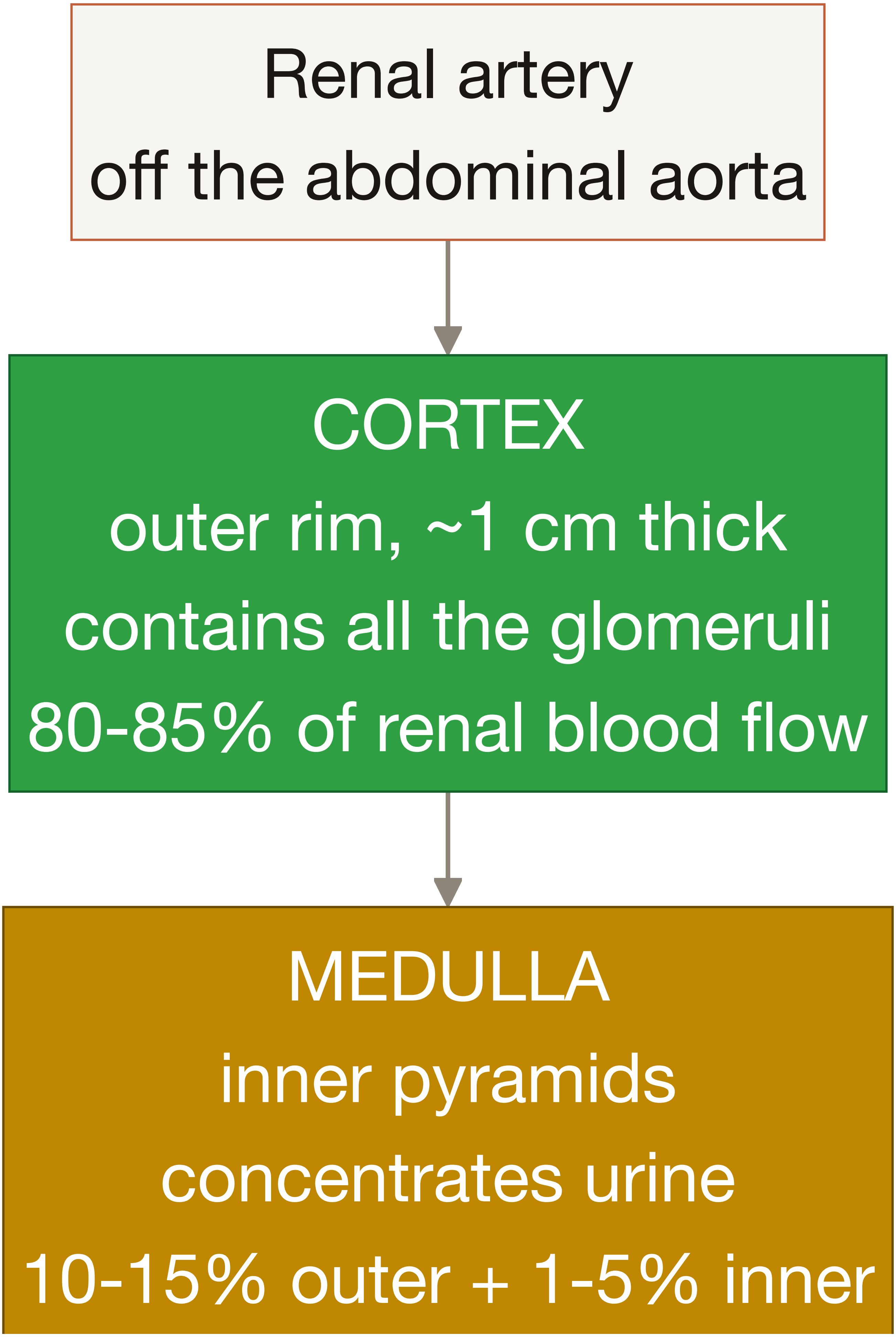
Measured values are somewhat lower than that review figure but still comfortably above brain:

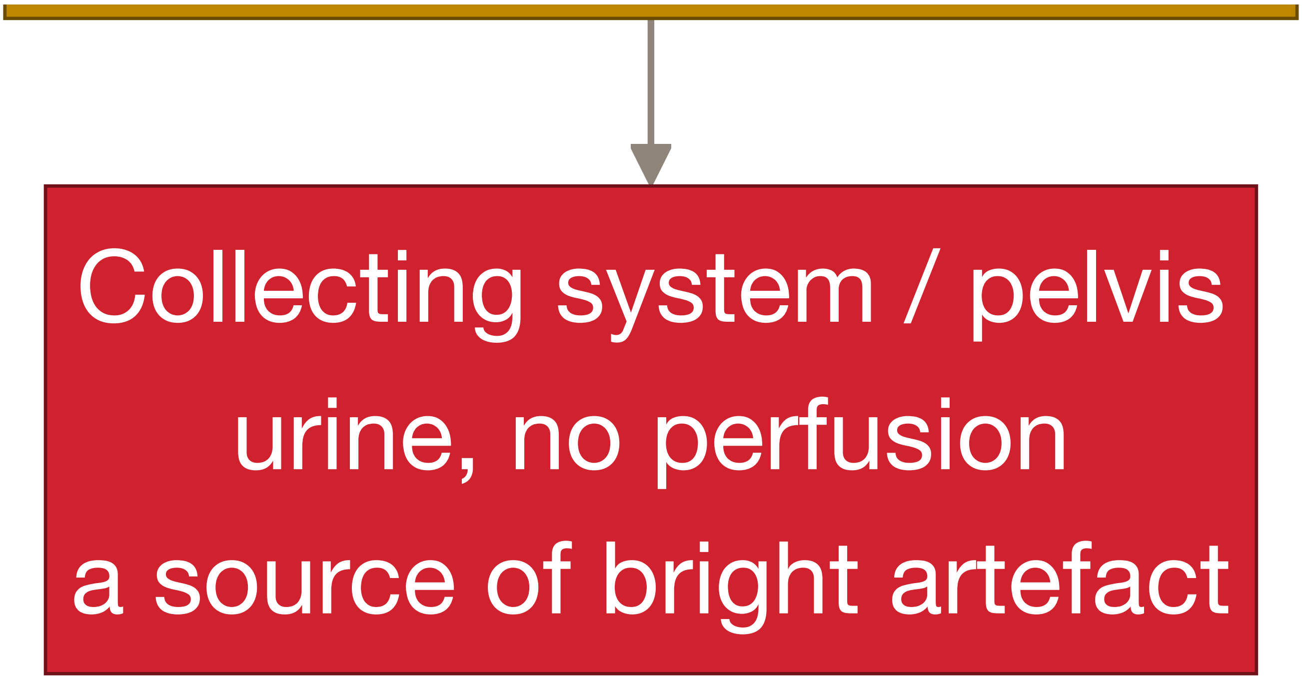
STUDY	ROI	PWS (% OF M0)	CONDITIONS
Garcia-Ruiz 2025	cortex	**2.95 ± 0.56%** (1.5T), **3.09 ± 0.59%** (3T)	n=16 healthy, PCASL SE-EPI, PLD 1.3 s, 2 BS pulses
Garcia-Ruiz 2025	medulla	1.36 ± 0.15% / 1.40 ± 0.19%	same
Bones 2021	cortex	**5.98 ± 0.70%** at peak TI ~1200 ms	n=7 healthy 23–34, 1.5T FAIR
Bones 2021	cortex	3.35 ± 0.83% at ~800 ms	n=7, 1.5T VSASL10
Bones 2019	whole kidney	2.07 / 2.16% free-breathing; 2.14 / 2.26% paced	n=9, 1.5T pCASL, **under heavy BGS2H only**
Radovic 2022	whole kidney (allograft)	3.43 ± 1.43%	n=18 paediatric/young-adult transplant, 1.5T FAIR, TI 2000 ms
Harteveld 2022	contralateral normal kidney	1.5 ± 0.61% @ PLD 0.5 s → 0.89 ± 0.39% @ 1.0 s → **0.51 ± 0.27%** @ 1.5 s	n=10 children mean 4.3 y, 1.5T pCASL

⚠️ **PWS is not transferable across protocols.** Those seven rows span 0.51% to 5.98% for the same organ, driven by TI/PLD, labelling scheme and background-suppression pulse count. Use PWS as a sanity check on subtraction polarity, never as a fixed SNR-style threshold.

PART 2 — The anatomy that matters

2.1 Cortex and medulla, in 90 seconds


Renal artery
off the abdominal aorta
CORTEX
outer rim, ~1 cm thick
contains all the glomeruli
80-85% of renal blood flow
MEDULLA
inner pyramids
concentrates urine
10-15% outer + 1-5% inner



The flow-share figures come from the NCBI Bookshelf preclinical renal ASL chapter: "*it is roughly estimated that 80–85% of the total renal blood flow supplies renal cortex, 10–15% outer medulla and only 1–5% inner medulla.*" Note these are **shares of total flow**, not per-gram perfusion — they do not translate into a cortex:medulla ratio without a mass correction, and the source itself calls them "roughly estimated".

The physiological anchor for magnitude, from Odudu 2018:

> *"All blood flow to the kidney (~25% of cardiac output at rest or ~1200 mL/min or ~400 mL/100 g/min in a 70 kg adult with a 300 g kidney) passes through the glomeruli of the renal cortices."*

So **~400 mL/100 g/min whole-kidney** is a useful outer sanity bound: a cortical value materially above it implies more blood than the kidney receives.

2.2 The cortico-medullary perfusion gradient — and why it is *not* a GM/WM analogue

The temptation is obvious: brain has GM ~2.5× WM, kidney has cortex > medulla, so port the GM/WM ratio check. **Do not.** Three reasons.

Reason 1 — the measured ratio is not stable. Computed from published means (this is my own arithmetic, not a published statistic):

STUDY	CORTEX	MEDULLA	RATIO	PROTOCOL
Haddock 2019	273 ± 38	**38 ± 5**	**7.2**	n=9 healthy 20–48, 3T FAIR bFFE, medulla ROI = **innermost half** of the medulla volume
Li LP 2016 (healthy)	207.3 ± 41.8	42.6 ± 15.8	4.9	n=30 HC mean 42.0 ± 18.1, 3.0T FAIR True-FISP

STUDY	CORTEX	MEDULLA	RATIO	PROTOCOL
Harteveld 2020 (FAIR)	362 ± 57	140 ± 47	2.6	n=15 healthy 51 ± 10, 3T
Harteveld 2020 (pCASL)	201 ± 72	84 ± 27	2.4	same subjects
Garcia-Ruiz 2025 (3T)	310.63 ± 52.72	133.83 ± 19.68	2.3	n=16 healthy 41.8 ± 13.3, PCASL multi-delay
Zhang 2024 (healthy)	317.59 ± 14.02	202.85 ± 28.50	1.6	n=10 HV, 3T 3D pCASL
Hammon 2016	337.10 ± 34.83	**279.61 ± 26.73**	**1.2**	n=14 (5 HV + 9 hypertensives) 48 ± 13, 1.5T FAIR True-FISP, 18-s breath-hold
Shirvani 2019	190.72 ± 32.35	~168–183	**1.1**	n=12 healthy, 3T FAIR 3D-GRASE, cortex = **outer 3 voxels**

That is 1.1 to 7.2 — a factor of **6.5**, driven almost entirely by **how the medulla mask was drawn**, not by physiology.

Reason 2 — the low end is a documented artefact, and one paper says so itself. Hammon 2016:

> *"the segmentation of the medulla may contain parts of the cortex what explains higher than expected medullary perfusion values"*

Hammon 2016 whole/cortex/medulla values and reproducibility

There is also an internal tell you can check without trusting anyone's prose: Hammon's **whole-kidney value (307.26 ± 25.65) sits *between* cortex (337.10) and medulla (279.61)**, with only 58 units separating the two compartments. That is the signature of a k-means split of a nearly uniform map, not a real gradient. Shirvani 2019 has a matching tell — their fitted tissue T1 is **799.61 ms cortex vs 807.11 ms medulla at 3T**, whereas Gillis 2014 measured renal medulla T1 at **1651 ± 86 ms at 3T**. Their segmentation almost certainly was not separating the compartments.

Reason 3 — the denominator is a quantity the field says is unreliable. Consensus **R10.2** (14% abstention, 89% agreement) declares medullary RBF not reliable, and Garcia-Ruiz measured medullary test–retest **ICC of 0.08 at 3T** (multi-delay). Building your discriminator on the least reliable measurement in renal ASL is self-defeating.

> ✅ **What the ratio *is* good for:** an **UNCALIBRATED segmentation-integrity flag**. A ratio near 1 is a reliable signature of a broken cortico-medullary segmentation. Report it against the *segmentation*, never as a FAIL on the CBF map.

2.3 The two kidneys move independently

A brain is one rigid body. Two kidneys are not. The consensus is explicit:

> "*Note that the kidneys should be registered separately if using rigid/affine transformations as they move independently.*" — Nery 2020, in the **R8.1** discussion

This is corroborated by three independent implementations: Bones 2019 uses a per-kidney translation- only 3D Euler transform in Elastix; Bones 2022 registers *per subject, per kidney, per slice*; Olsen 2025 applies rigid-body affine "*separately for each repetition and kidney*".

Combined with **R10.1** (report left and right separately, 100% agreement), this changes the data model: **a renal QC report has at minimum two ROIs per subject, not one.**

2.4 Transplants are a different organ, geometrically

A transplanted kidney sits in the **iliac fossa**, anastomosed to the iliac artery. Consequences, all from the consensus's Transplants section:

- Much less respiratory motion — "*making respiratory gating/triggering less important for ASL in the transplant setting.*"
- PCASL "*may be preferable*" (the consensus's words — not "preferred") because of the flexibility in positioning the labelling and imaging planes.
- Labelling plane goes perpendicular to the abdominal aorta **just above the iliac bifurcation**.
- Baseline iliac flow "*likely changes the absolute perfusion of the transplanted kidney compared to the native kidney, biasing absolute perfusion towards lower values.*"

And a practical consequence measured by Echeverria-Chasco 2023 (n=20 enrolled / 18 ASL analysed, 54 ± 14 y, 120 ± 105 months post-transplant, eGFR 76.89 ± 13.96, 3.0T Siemens Skyra, pCASL SE-EPI, PLD 1.2 s):

> "*Because the allograft is placed in the iliac fossa under different orientations, orienting the imaging slab along the long axis of the kidney was not always possible.*"

So the **coronal-oblique orientation check must be transplant-aware**, or it will fail every allograft.

2.5 The renal-specific artefact sources with no brain analogue

SOURCE	EVIDENCE
Air in lungs and bowel → susceptibility distortion	de Boer 2021 (n=19, 3T FAIR GE-EPI): "*Artifacts were limited to geometrical distortion due to B0 inhomogeneities and susceptibility effects due to air in the lungs and digestive tract, which did not affect the perfusion quantification.*" — a WARN, not a FAIL
Intestinal peristalsis	Radovic 2022: "*Intestinal peristalsis was found to be a possible source of motion artefacts, which was resolved along with other important clinical factors by organising patients to come early in the morning fasted.*"
The collecting system / renal pelvis	Gillis 2014 abandoned medullary perfusion entirely: "*Perfusion maps generated via post processing result in a heterogeneous appearance of the renal medulla, probably due to the presence of larger vessels and the renal pelvis.*" Franklin 2021 saw "*small abnormalities at locations that are part of the collecting system*"

SOURCE	EVIDENCE
Perirenal fat	Bones 2019: "*fatty tissue is unaffected by contrast agents and recovers quickly from preceding BGS pulses due to the short T1*" — fat survives background suppression, which is why **R7.6** recommends fat suppression (5% abstention, 90% agreement)
Chemical shift	Consensus: fat suppression is "*particularly important for readout schemes with a low phase-encoding bandwidth, such as the aforementioned recommended EPI readout*"
bSSFP banding	Consensus: "*A bSSFP scheme is also sensitive to field inhomogeneity, which results in banding artefacts in areas of off-resonance within the image.*" Brain ASL almost never uses bSSFP; renal FAIR frequently does
Anatomical un-plannability	de Boer 2021: "*In one subject, FAIR-ASL could not be planned because the kidneys were located in the same coronal plane as the aorta.*" (1 of 19)

PART 3 — Brain vs kidney, difference by difference

This is the table to keep open while porting a check. Brain column = the 2015 Alsop White Paper (`../researchpaper/nihms591163.pdf`). Kidney column = Nery 2020 unless stated.

#	ASPECT	🧠 BRAIN (ALSOP 2015)	🫘 KIDNEY (NERY 2020)	CONSEQUENCE FOR OSIPY-QC
1	**Labelling scheme**	PCASL is *the* recommendation	**Both FAIR (PASL) and PCASL adequate** (R3.1, 93%)	Cannot assume one. Every check branches
2	**PASL variant**	FAIR named as one of several	**FAIR is *the* renal PASL variant**: "*The FAIR variant is the PASL scheme most widely used in renal imaging*" (Odudu 2018)	`PASLType: "FAIR"` is the common path, not an edge case
3	**Labelling geometry**	Plane ~85 mm below AC–PC	PCASL: perpendicular to **aorta**, **8–10 cm superior** to kidney centre (R5.3 100%, R5.4 94%). FAIR: selective slab **excludes the aorta** (R4.2, 100%), thickness = imaging slab + **10–20 mm** (R4.3, 86%)	Two mutually exclusive geometry checks keyed off detected scheme
4	**Labelling duration**	PCASL 1800 ms	PCASL **1.5–1.8 s** (R5.2, 100%)	Overlapping but wider
5	**PLD / TI**	PLD 1800 ms adults <70; 2000 ms >70 and clinical	**PCASL PLD 1.2–1.5 s** (R5.11, 100%); **FAIR TI 1.8–2.0 s** (R4.4, 89%), TI1 **1.0–1.2 s** (R4.6, 92%)	Renal PLD is **shorter** than brain. A brain-tuned "PLD too short" rule would fire on every good renal scan
6	**Bolus control**	QUIPSS II, TI1 = 800 ms	**QUIPSS II or Q2TIPS mandatory** for FAIR quantification (R4.5, 25% abstention, **100%** agreement)	Hard check on the FAIR path
7	**Averages**	not specified numerically	**≥20 ASL pairs** (R4.7 89% / R5.12 83%) — but the body text scopes this to "*when using the recommended 2D readout*"	**Gate the check on readout type**, or it mis-fires on 3D
8	**TR**	—	**4–6 s** including labelling + readout (R6.12, 0% abstention, 94%)	Clean metadata check
9	**Readout**	**3D** segmented RARE/GRASE recommended	**2D single-slice** default (R6.1, 95%); 2D multislice for extended coverage (R6.2, 100%); **3D explicitly not the default** (R6.3, 95%)	Inverted. The default artefact set is EPI distortion + chemical shift, not 3D blurring

#	ASPECT	🧠 BRAIN (ALSOP 2015)	KIDNEY (NERY 2020)	CONSEQUENCE FOR OSIPY-QC
10	**Pulse sequence**	EPI or spiral	**Spin-echo EPI preferred** (R6.4, **75%** — the lowest passing agreement in the document, and scoped to *2D single-slice*); bSSFP and single-shot RARE adequate alternatives (R6.5, 94%). Gradient-echo EPI was excluded from consideration	Readout-conditioned artefact expectations
11	**Orientation**	axial	**Coronal oblique along the kidney long axis** (R6.7, 93%) — for two reasons: FAIR needs it to avoid labelling inflowing vessels, *and* it puts respiratory motion **in-plane** so 2D registration can fix it	Checkable from the NIfTI affine. Must be transplant-aware
12	**Voxel size**	3–4 mm in-plane, 4–8 mm through	in-plane **2–4 mm** (R6.10, 93%); slice **4–8 mm** 2D (R6.8, 100%) / **3–6 mm** 3D (R6.9, 100%)	Direct header check
13	**Acceleration**	—	up to **R = 2** (R6.11, 100%) — permissive wording ("*may be used*"), so WARN above, not FAIL	Metadata check
14	**Breathing**	n/a — the head is still	**Breath-hold NOT recommended** (R7.3, 94%); **free breathing preferred** (R7.4, **76%** — second-weakest); **respiratory triggering advantageous** (R7.5, 95%)	Free breathing + retrospective correction is the *expected* case
15	**Background suppression**	recommended for ASL pairs	recommended for ASL pairs (R7.2, **80%**), typically **2–4** adiabatic inversion pulses	**Same direction, not opposite.** See #16
16	**M0 and BS**	M0 must have BS **OFF**	*"The PD image should be acquired without labelling or BGS"*	**Identical rule.** Do not invert your existing Module 6
17	**M0 requirement**	*recommended*	**mandatory** (R9.1, 0% abstention, 94%)	Renal missing-M0 can legitimately be a harder failure than brain
18	**M0 TR**	>5 s, else ×1/(1−e^(−TR/$T1_{tissue}$))	**same rule, same 5 s**, same correction — but $T1_{tissue}$ is *kidney* T1, and the consensus gives no value for it	Port the check; source the constant separately
19	**Fat suppression**	not a brain issue	**recommended** (R7.6, 90%)	New check, metadata-only
20	**Registration**	rigid, one brain	**highly recommended** (R8.1, 13% abstention, **100%**), **each kidney separately**	Per-kidney, not per-volume

#	ASPECT	🧠 BRAIN (ALSOP 2015)	🫘 KIDNEY (NERY 2020)	CONSEQUENCE FOR OSIPY-QC
21	**Outlier rejection**	SCORE/SCRUB/ENABLE	**recommended** (R8.2, 0% abstention, **100%**) — with **no method and no threshold specified**	The clearest opening for the toolbox
22	**Tissue priors**	GM/WM/CSF probability maps	**none.** Manual ROI is the default (R9.10, 100%)	**QEI does not port as written**
23	**Reported ROI**	whole-brain GM	**cortex, left and right separately** (R10.1, 0% abstention, **100%**); whole-kidney explicitly rejected	Two verdicts per subject
24	**Medulla / WM analogue**	WM is a valid ROI	**medullary RBF "not considered reliable"** (R10.2, 89%)	Medulla → INFO/N-A, never PASS/FAIL
25	**λ**	0.9 mL/g	**0.9 mL/g** (R9.4, 90%) — but **borrowed from brain**, see §4.3	Same number, weaker evidence
26	**$T1_{blood}$ @3T**	1650 ms	**1.65 s — identical** (R9.5, 0% abstention, 100%)	No change
27	**$T1_{blood}$ @1.5T**	1350 ms	**1.48 s — differs** (R9.6, 93%)	Change on the 1.5T path
28	**α PCASL**	0.85	**0.85 — identical** (R9.8, 86%)	No change
29	**α PASL**	0.98	**0.95 — differs** (R9.7, 100%)	Change on the FAIR path
30	**BS efficiency penalty**	~5%/pulse noted	**×0.93 per BGS inversion pulse** (R9.9, 19% abstention, 100%)	Fully computable. e.g. PCASL + 3 BS pulses → $0.85 \times 0.93^3 = \mathbf{0.684}$
31	**Quantification model**	single-compartment	**single-compartment** (R9.2, 100%); two-compartment **not** the default (R9.3, 95%)	Same
32	**Multi-delay**	optional	**single time point recommended** (R3.2, 86%); multi useful "*if delayed arrival time is suspected*" (R3.3, 100%). Odudu 2018: multi-delay "*cannot be currently universally recommended*"	Single-delay is the expected input
33	**Standard data layout**	ASL-BIDS	**none.** ASL-BIDS says: "*ASL-BIDS is validated in ASL images of the brain only*"	See §7.6
34	**Reference pipeline**	ASLPrep, ExploreASL, BASIL	**none.** No renal ASL pipeline exists	Cannot consume pipeline derivatives
35	**Published quality index**	QEI (Dolui 2024), cutoff ~0.5	**none**	§7.1

> 💡 **The three that will trip you up most:** #9 (2D not 3D), #16 (the M0/BS rule is the **same**, not inverted — an easy and embarrassing mistake), and #22 (no tissue priors, therefore no QEI).

PART 4 — The consensus, quoted

4.1 What Nery 2020 actually is

- **Title:** *Consensus-based technical recommendations for clinical translation of renal ASL MRI*
- **Journal:** MAGMA 33(1):141–161, DOI `10.1007/s10334-019-00800-z`, **CC-BY**
- **Process:** modified Delphi — online surveys plus two in-person meetings
- **Panel: 23 renal ASL experts** (the paper has 26 authors). Composition derived from Table 1: 16 physicists, 4 engineers, 2 radiologists, 1 nephrologist, across 6 countries
- **Threshold:** "*Consensus was assumed to be found when 75% or more of respondents agreed.*" **Neutral responses counted as abstentions and were excluded from the denominator** — this is why agreement + abstention can exceed 100%. Quote that mechanic or a reviewer will think you made an arithmetic error
- **Output: 59 statements achieved consensus**; **2 failed**, both on patient preparation
- **Funding:** EU COST Action **CA16103 (PARENCHIMA)**

Nery 2020 Table 2, part 1 — the numbered consensus statements

Nery 2020 Table 2, part 2 — preprocessing, quantification and reporting statements

> ⚠️ **Correction to a number that circulated in my working notes:** this paper has **59** consensus statements, not 166. Page 6 says so directly: "*The 59 final consensus statements are listed in Table 2.*"

4.2 The statements that become Stream-A checks

Format: **R-number (abstention %, agreement %)**. All verbatim.

Field strength and scheme

- **R2.1** (0, 87) — "*Both 1.5 T and 3 T are adequate field strengths*"
- **R3.1** (6, 93) — "*Both PASL:FAIR and PCASL are adequate labeling strategies*"
- **R3.2** (6, 86) — "*Single time point acquisitions are recommended for simplicity of acquisition and data analysis*"
- **R3.3** (6, 100) — multi-timepoint "*can be useful if delayed arrival time is suspected in a clinical population*"

FAIR path

- **R4.2** (6, 100) — "*The selective slab should be carefully positioned, excluding the aorta*"
- **R4.3** (13, 86) — "*The selective inversion slab thickness should equal the imaging slab thickness + [10–20] mm*"
- **R4.4** (10, 89) — "*In single-TI acquisitions, an inversion time of 1.8–2.0 s is recommended*"
- **R4.5** (25, 100) — "*In single-TI acquisitions, an approach for controlling the temporal width of the bolus (QUIPSS II or Q2TIPS) must be used to quantify perfusion*"
- **R4.6** (25, 92) — "*A bolus duration (TI1) of 1.0–1.2 s is recommended*"

- **R4.7** (10, 89) — "*In single-TI acquisitions, a minimum of 20 ASL pairs is recommended*"

PCASL path

- **R5.3** (13, 100) — "*The labeling plane should be oriented approximately perpendicular to the aorta*"
- **R5.4** (14, 94) — "*The labeling plane should be positioned at approximately 8–10 cm from the centre of the kidney, in the superior direction*"
- **R5.2** (10, 100) — "*A labeling time of 1.5–1.8 s is recommended*"
- **R5.11** (19, 100) — "*In single PLD acquisitions, a PLD of 1.2–1.5 s is recommended*"
- **R5.12** (14, 83) — "*In single-PLD acquisitions, a minimum of 20 ASL pairs is recommended*"
- **R5.8** (27, 100) — "*Average B1 of 1.6 μT is recommended*"; **R5.9** (25, 92) — Gave 0.4–0.6 mT/m

Readout and geometry

- **R6.1** (10, 95) — 2D single-slice default; **R6.3** (10, 95) — 3D not the default
- **R6.4** (5, **75**) — "*Spin-echo EPI is the preferred readout for 2D single-slice acquisitions*"
- **R6.7** (6, 93) — "*Coronal oblique slices (along the major axis of the kidneys) are preferable for renal ASL*"
- **R6.8** (19, 100) 2D slice 4–8 mm · **R6.9** (13, 100) 3D slice 3–6 mm · **R6.10** (0, 93) in-plane 2–4 mm
- **R6.11** (19, 100) — acceleration "*up to R = 2 may be used*"
- **R6.12** (0, 94) — "*The recommended TR (including labeling + readout) is 4–6 s*"

Motion and preprocessing — the QC block

- **R7.2** (5, 80) — "*Background-suppression is recommended for renal ASL*"
- **R7.3** (0, 94) — "*Breath-hold scans are not recommended for clinical renal ASL*"
- **R7.4** (0, **76**) — "*Renal ASL scans under free breathing are preferred*"
- **R7.5** (5, 95) — "*Respiratory triggering can be advantageous to minimize the effects of kidney motion at the expense of scan time*"
- **R7.6** (5, 90) — "*Fat suppression is recommended for renal ASL*"
- **R8.1** (13, **100**) — "*Retrospective image registration is highly recommended for renal ASL*"
- **R8.2** (0, **100**) — "*Outlier rejection is recommended for renal ASL*"

Reporting

- **R9.10** (0, 100) — "*Regions of interest selection should be performed manually as the default approach. Semi-automatic methods may be used if local expertise is available (e.g. using T1 maps) but require extensive validation*"
- **R9.11** (6, 93) — ROI "*based on the ASL M0 image or a separately acquired structural dataset*"
- **R10.1** (0, **100**) — "*Cortical renal blood flow values (not whole-kidney) should be reported, separately for left and right kidney*"
- **R10.2** (14, 89) — "*Medullary renal blood flow values are not considered reliable with current measurement approaches*"

> ⚠️ **Watch the weak ones.** R7.4 (76%) and R6.4 (75%) barely cleared the 75% bar. R5.10 (Gmax/Gave 6–7) carries **40% abstention** — the highest in the document — *and* the discussion text says "a ratio of 7" while Table 2 says "6–7". Anything built on R5.10 is advisory at best.

4.3 The quantification block

Nery 2020 — the RBF equations for PCASL and FAIR

PCASL:

```
RBF = 6000 · λ · (SI_control − SI_label) · exp(PLD / T1_blood)
      ─────────────────────────────────────────────────────────     [mL/100 g/min]
      2 · α · T1_blood · SI_PD · (1 − exp(−τ / T1_blood))
```

FAIR with QUIPSS II / Q2TIPS bolus saturation:

```
RBF = 6000 · λ · (SI_control − SI_label) · exp(TI / T1_blood)
      ───────────────────────────────────────────────────────     [mL/100 g/min]
      2 · α · TI1 · SI_PD
```

Constants (all with their agreement levels):

CONSTANT	RENAL VALUE	BRAIN WHITE PAPER	SAME?
λ (partition coefficient)	**0.9 mL/g** (R9.4, 5/90)	0.9 mL/g	✅ same number
$T1_{blood}$ @ 3T	**1.65 s** (R9.5, 0/100)	1650 ms	✅ identical
$T1_{blood}$ @ 1.5T	**1.48 s** (R9.6, 13/93)	1350 ms	❌ **differs**
α PASL	**95%** (R9.7, 6/100)	0.98	❌ **differs**
α PCASL	**85%** (R9.8, 13/86)	0.85	✅ identical
BS penalty	**×0.93 per inversion pulse** (R9.9, 19/100)	~5%/pulse noted	≈ same

Both α statements are explicitly parenthesised "*(neglecting background suppression loss)*", and R7.2 recommends BS. **So the recommended renal protocol is exactly the case where the raw α values must not be used.** A quantifier that hard-codes 0.85 or 0.95 will be systematically wrong on the recommended acquisition.

> 🚩 **The λ caveat you must carry.** The consensus says so itself:
>
> *"Since a reliable reference for the partition coefficient value in kidney was not known to this group, we recommend the use of a value of 0.9 mL/g [R9.4], the average value for brain tissue. Indeed, literature on water content in the renal cortex suggests the value of 0.9 mL/g to be a good approximation … Since this is a constant factor across the image, perfusion values calculated with this assumption could be readily corrected when a more accurate value of λ is known."*
>
> λ = 0.9 is a **consensus-endorsed borrowed brain constant**, not a renal measurement. Tag it `published-as-consensus-assumption`, never `published-as-measurement`. For context, the field uses 0.8 (Gillis 2014, Cox 2017), 0.9 (consensus, Harteveld, Shirvani), 0.91 (measured in **dog** kidney by Hillaert 2024 from a water content of ~79–81% and blood density 1.05 g/mL), and the preclinical literature spans **0.52–0.94** with the Chuang chapter stating flatly that "*the suitable value for the kidney remains to be determined.*"
>
> ⚠️ **Provenance depth warning:** the two blood-T1 values trace to a **single** primary source — Zhang et al., MRM 2013;70:1082–1086 (reference [98] of the consensus), which is **paywalled and not in my local library**. Until it is read, tag both as `published-via-consensus`, not `published-at-primary-level`.

4.4 The M0 rule — the cleanest port from brain

> *"we consider acquisition of a separate PD image (also referred to as M0 image) a mandatory step for ASL quantification [R.9.1]. The PD image should be acquired without labelling or BGS and using a similar readout and acquisition parameters, with the exception that a long TR should be used. If this image is acquired without waiting for a sufficiently (> 5 s) long recovery time (TR), the SI_PD should be corrected for incomplete relaxation using the equation:*
>
> `SI_PD,corr = SI_PD / (1 − exp(-TR / T1_tissue))`
>
> *where T1_tissue is an estimate of the kidney T1."*

Compare the brain White Paper: "*If TR is less than 5 s, the PD image should be multiplied by the factor (1/(1 − e^(−TR/T1,tissue))), where T1,tissue is the assumed T1 of gray matter.*" **Same threshold, same functional form, same direction.** Only the tissue T1 changes.

> ⚠️ **But the kidney has two tissue T1s, and the consensus gives neither.** I searched the entire paper with a regex for any `\d{3,4} ms` pattern: **zero hits**. The only mention is "*where T1_tissue is an estimate of the kidney T1.*" The paper mandates a correction it does not parameterise. Published healthy values must come from elsewhere — see §5.6.

4.5 The two statements that failed

Both concern patient preparation, and both failed hard:

FAILED STATEMENT	ABSTENTION	AGREEMENT
"Diet needs to be controlled before the scan"	43%	50%

FAILED STATEMENT	ABSTENTION	AGREEMENT
"Subjects are required to follow a controlled and standardized salt intake before the scan"	48%	27%

Only **R1.1** (24% abstention, 100% agreement) passed on preparation: "*Subject should be scanned in a normal hydration status when clinically appropriate.*"

4.6 What the consensus does NOT contain

This is the most decision-relevant finding in the whole document, and it is a **negative**:

- ❌ No numeric image-quality threshold of any kind — no tSNR floor, no CoV cutoff, no motion limit, no negative-voxel fraction, no SNR floor, no renal QEI.
- ❌ No reference perfusion values. `mL/100` appears exactly **once** in the paper, in the phrase "*typically mL/100 g/min*", with no number attached.
- ❌ No kidney tissue T1 value (§4.4).
- ❌ No outlier-rejection method or criterion, despite R8.2 being unanimous.
- ❌ No paediatric or age-specific recommendation anywhere among the 59 statements. *(This is established by exhaustive search of the document, not by any sentence in it — an absence cannot be quoted.)*
- ❌ No mention of vascular crusher gradients or flow suppression, unlike the brain White Paper.

The panel is candid about why:

> *"Another limitation of this work is that experimental data were lacking to provide a definitive answer for certain issues. When no published data were available to support a specific statement or the existing evidence was not strong, members were asked to provide an answer based on their expert opinion. The consensus protocol proposed here can thus be considered a starting point that will likely be modified in the future as more data become available."*

> ✅ **Quote that paragraph in the design doc.** It is the strongest published justification you will find for tagging every renal quality threshold `uncalibrated`.

Nery 2020 Table 4 — minimum set of parameters to be reported

Table 4 is the closest thing to a renal sidecar schema and is agreed at **81–100%**:

- **General MR:** scanner manufacturer/model, receive coil type, pixel bandwidth, fat suppression, field of view, magnetic field strength, flip angle, image orientation, in-plane resolution, number of slices, parallel imaging (technique + acceleration factor), partial Fourier, physiological triggering/gating, readout pulse sequence type, slice gap, slice ordering, slice thickness, echo time, repetition time.
- **ASL-specific:** background suppression, inflow time(s)/post-labelling delay(s), labelling duration, labelling type, number of averages, quantification model.
- Plus, from the discussion: "*a minimum of mean and standard deviation of cortical renal ASL perfusion values should be reported at the subject (i.e. ROI analysis) or group level. The median should also be considered in the presence of skewed RBF distributions.*"

Roughly **eight** of those items have **no BIDS field at all** — most importantly physiological triggering/gating, fat suppression, pixel bandwidth and slice gap. See §7.6.

PART 5 — Normal values, with everything attached

5.1 Read this before you use any number below

> 🛑 **Every value in this section is a COHORT MEAN, not a threshold.** Not one paper in my 51-PDF library proposes a PASS/FAIL band for renal perfusion. Converting a mean ± SD into a band is the single most likely way to ship something indefensible.

Four technical confounds move renal cortical perfusion **before any physiology**:

CONFOUND	EFFECT SIZE	EVIDENCE
Labelling scheme	**~80%**	FAIR 362 ± 57 vs pCASL 201 ± 72, **same 15 subjects, same session** (Harteveld 2020, 3T)
Field strength	**~11%**	1.5T 347.65 ± 54.08 vs 3T 310.63 ± 52.72, **same 16 subjects, same day** (Garcia-Ruiz 2025, multi-delay PCASL, p<0.001)
Age	**~20%**	<40 y 383.90 ± 61.80 vs ≥40 y 306.10 ± 41.33 at 1.5T single-delay (Garcia-Ruiz 2025, n=8 vs 8, age effect p=0.005)
ROI definition	**~7%**	Two competent human observers differ by 14 mL/100 g/min on the same data (Bones 2022, 1.5T pCASL)

Garcia-Ruiz 2025 — RBF and T1 by field strength and age

Note the field-strength direction is **counter-intuitive**: higher field gives *lower* measured RBF. That alone is a good reason to store field strength as an explicit band key rather than assume it.

Not a confound: sex. Garcia-Ruiz found no significant sex effect on RBF (p = 0.076 cortex, p = 0.621 medulla) — but ATT *was* significantly shorter in females. Stratify an ATT check by sex; do not stratify a perfusion band by it.

5.2 Cortical perfusion — the healthy adult picture

The literature-wide envelope (Odudu et al. 2018, systematic review of **53 studies** published in English through January 2018; studies "*generally small (mean 25 ± 23 participants, range 4–98)*", mixed 1.5T/3T, mixed FAIR/pCASL):

> *"Renal cortical perfusion by ASL ranged from 139 to 427 mL/100 g/min in healthy volunteers and from 83 to 412 mL/100 g/min in a broad range of patient groups."*

Odudu 2018 — the reported perfusion ranges

⚠️ **Three reasons that range is not a PASS band.** (1) It is a spread of **study-level means**, not of individuals — mechanically narrower than the individual distribution. (2) Healthy (139–427) and patient (83–412) ranges **overlap almost entirely**, so it has essentially no discriminative power. (3) I could not trace the lower bound of 139 to any row of the review's own supplementary Table S1 — the lowest healthy-control cortical mean in that spreadsheet is **117 ± 24** (Shimizu 2017, older subgroup). The 427 upper bound *does* trace, to Artz 2011 (eGFR>60 subgroup, 427 ± 20).

Individual-subject spread, which is what you would actually need: Olsen et al. 2025 report **150–422 mL/min/100 mL** for ASL across 10 healthy subjects (mean age 25.2 ± 3.2, 5 male, 3T GE Signa PET/MR, multislice FAIR + background suppression, single-shot SE-EPI, TI 1500 ms, 20 pairs). Note the units — **per 100 mL, not per 100 g**.

Study-by-study, healthy adult native kidneys, cortex:

VALUE (ML/100 G/MIN)	N	AGE	FIELD	LABELLING	SOURCE
441 ± 84	7	23–34	1.5T	FAIR	Bones 2021
383.90 ± 61.80 (<40 y)	8	25–38	1.5T	PCASL single-delay	Garcia-Ruiz 2025
362 ± 57	15	51 ± 10	3T	multi-delay FAIR, GE-EPI	Harteveld 2020
347.65 ± 54.08	16	41.8 ± 13.3	1.5T	multi-delay PCASL, SE-EPI	Garcia-Ruiz 2025
337.10 ± 34.83	14 (5 HV + 9 hypertensive)	48 ± 13	1.5T	FAIR True-FISP, 18-s breath-hold	Hammon 2016
321 → 334 (2 visits)	12 (24 kidneys)	44.1 ± 14.6	3.0T	FAIR True-FISP, TI 900 ms, λ=0.8	Gillis 2014
317.59 ± 14.02	10	not stated	3T	3D pCASL	Zhang 2024
310.63 ± 52.72	16	41.8 ± 13.3	3T	multi-delay PCASL	Garcia-Ruiz 2025
307.46 ± 90.53 (donating kidney)	10 living donors	46 ± 6	1.5T	FAIR 3D-GRASE, TI 2000 ms	Radovic 2025
295.09 ± 59.69	54	52.19 ± 10.27	3.0T	pCASL	Shi 2026 (MN controls)
295 ± 97 *(voxelwise SD)*	**5 adults**	32 ± 5	1.5T	FAIR Q2TIPS 3D-GRASE, TI 1200 ms	Nery 2019
283 ± 55	10 of 15	23–38	1.5T	balanced pCASL	Bones 2020
273 ± 38 (230–343)	9	20–48	3T	FAIR bFFE	Haddock 2019
271.8 ± 43.5	7	25 ± 3	3.0T	pCASL bSSFP	Ahn 2020 *(abstract only)*

VALUE (ML/100 G/MIN)	N	AGE	FIELD	LABELLING	SOURCE
255 ± 70	85 (of 127)	41 ± 19	**1.5T AND 3T pooled**	FAIR, λ=0.8, T1b 1.55/1.36 s	Cox 2017
221.62 ± 31.08 (single-TI)	12	27.6 ± 18.5	3T	FAIR 3D-GRASE, cortex = outer 3 voxels	Shirvani 2019
211 ± 31 (auto) / 208 ± 31 (manual)	10	23–38	1.5T	balanced pCASL	Bones 2022
207.3 ± 41.8	30	42.0 ± 18.1	3.0T	FAIR True-FISP	Li LP 2016
201 ± 72	15	51 ± 10	3T	multi-delay pCASL, GE-EPI	Harteveld 2020
201 ± 36 (SE-EPI)	10	27 ± 10	3T	FAIR, respiratory-triggered	Buchanan 2018
197	25	31 (22–67)	1.5T	multi-TI FAIR 3D-GRASE	Cutajar 2012 *(abstract only)*
139.10 ± 37.93 (ATT-corrected)	14 **males only**	27.0 / 64.8 subgroups	3.0T	pcASL, α assumed 0.75	Shimizu 2017

That is 139 to 441 — a factor of 3.2, in healthy adults. Every one of those is a legitimate published measurement. The spread is protocol, not biology.

> ⚠️ **Two rows need a footnote.** (1) Nery 2019's `295 ± 97` is a **voxelwise** SD across the cortical ROI, not a between-subject SD — the paper says "*Results … are expressed as mean ± SD considering all voxels within the cortical ROI.*" The between-subject spread is the **range 245–343**, from 5 adults in a best-case "still" acquisition. Never compute mean ± 2SD from it. (2) The `295 ± 97` figure is also frequently miscited as a **paediatric** value because the abstract says only "healthy volunteers" — the Methods say "*Five healthy adult volunteers (age [y] = 32 ± 5 … range = 26–37; 3 male).*"

5.3 Cortical perfusion in disease

The reason a low-perfusion gate cannot be a hard check:

COHORT	CORTICAL RBF	N	FIELD	LABELLING	SOURCE
Healthy	317.59 ± 14.02	10	3T	3D pCASL	Zhang 2024
CKD stage 1	300.88 ± 17.30	6	3T	3D pCASL	Zhang 2024
CKD stage 2	245.36 ± 29.43	7	"	"	"

COHORT	CORTICAL RBF	N	FIELD	LABELLING	SOURCE
CKD stage 3	182.66 ± 24.35	15	"	"	"
CKD stage 4	114.83 ± 20.58	9	"	"	"
CKD stage 5	**110.53 ± 25.85**	5	"	"	"
Healthy	207.3 ± 41.8	30	3.0T	FAIR True-FISP	Li LP 2016
Diabetic stage-3 CKD	**108.4 ± 36.4**	33	"	"	"
Healthy (Cox cohort)	255 ± 70	85	1.5T+3T	FAIR	Cox 2017
CKD	**83 ± 68**	11	"	"	"
Membranous nephropathy, moderate–severe	186.26 ± 34.27	11 (age **70.00 ± 5.76**)	3.0T	pCASL	Shi 2026
Renal artery stenosis, GFR ≥30	175.5 ± 76.6	118 kidneys / 62 pts, 63.3 ± 9.1	3.0T	pCASL, PLD 2000 ms	Zhao 2025
Renal artery stenosis, GFR <30	133.7 ± 45.2	"	"	"	"
Diabetic kidney disease, left posterior **segment**	65.75 ± 19.36	9	3.0T	3D-pCASL, PLD 700 ms	Fang 2026

🚩 **Look at what that does to a threshold.** Healthy cohorts run 139–441. Diseased cohort *means* reach 65–110, and individual patients go lower. Zhang's CKD5 mean minus 2 SD is ~59; Li's CKD group mean is 108. Meanwhile Fang's *healthy* segmental values reach as low as 104–133 because they used small segmental ROIs at a very short PLD (700 ms). **The healthy and diseased distributions overlap almost completely once protocol is included.**

The only defensible use of an absolute renal perfusion value in v1.0 is an **implausibility guard** — flag 900, flag 5, say nothing about 150 — reported as **INFO/WARN**, never FAIL.

⚠️ **A miscitation to avoid:** the Kidney Int Rep CKD paper (PMC5575771) is **Li L-P, Tan H, Thacker JM, Li W, Zhou Y, Kohn O, Sprague SM, Prasad PV** — *not* Cox et al., as my working notes originally had it. The numbers were right; the attribution was to an entirely different group. Note also its controls were scanned at **TI 1.5 s** and patients at **TI 2.0 s** — the paper itself concedes "*Although this is not ideal*" — and the groups differ by 26 years in age.

5.4 Medullary perfusion — and why you should not grade it

VALUE (ML/100 G/MIN)	N	FIELD	LABELLING / MEDULLA ROI	SOURCE
38 ± 5 (34–50)	9 healthy	3T	FAIR bFFE, ROI = **innermost half** of the medulla volume	Haddock 2019
42.6 ± 15.8	30 healthy	3.0T	FAIR True-FISP	Li LP 2016
84 ± 27	15 healthy	3T	multi-delay pCASL	Harteveld 2020
133.83 ± 19.68	16 healthy	3T	multi-delay PCASL	Garcia-Ruiz 2025
140 ± 47	15 healthy	3T	multi-delay FAIR	Harteveld 2020
153.64 ± 25.46	16 healthy	1.5T	multi-delay PCASL	Garcia-Ruiz 2025
202.85 ± 28.50	10 healthy	3T	3D pCASL	Zhang 2024
279.61 ± 26.73 ⚠️	14	1.5T	FAIR True-FISP, authors admit cortical contamination	Hammon 2016

38 to 280 — a factor of **7.4**, for the same tissue. The consensus says why:

> *"Medullary perfusion is difficult to measure reliably … because of its lower perfusion (and, therefore, lower signal) and close proximity to cortex which makes it susceptible to partial volume contamination. The kinetics of labelled water are also uncertain because arterial water is divided between filtrate and smaller arterioles."*

And the reproducibility data backs it up hard (Garcia-Ruiz 2025, test–retest, n=14 completed):

ROI	FIELD / METHOD	WSCV	ICC
Cortex	1.5T multi-delay	**7.27%**	0.83 [0.55–0.94]
Cortex	3T multi-delay	10.50%	0.72 [0.34–0.90]
Medulla	1.5T multi-delay	15.78%	**0.43** [−0.12–0.77]
Medulla	**3T multi-delay**	**19.38%**	**0.08** [−0.45–0.57]

All four medullary ICC confidence intervals include zero. An ICC of 0.08 means essentially no subject-specific information survives between sessions.

> ✅ **Design decision:** medullary metrics are **INFO / N-A**, never PASS/WARN/FAIL. That is not conservatism — it follows directly from R10.2 plus two independent measurements. Two careful groups reached the same conclusion in practice: Gillis 2014 fell back to whole-kidney perfusion, and de Boer 2021 wrote "*In accordance with a recent recommendation article, medullary perfusion was not reported for FAIR-ASL since it was deemed unreliable due to low SNR.*"

5.5 Arterial transit time

Renal ATT is short compared to brain — but it is strongly method-dependent, and it does not reproduce well.

COHORT	CORTEX ATT	MEDULLA ATT	PROTOCOL
Bones 2021, n=7 healthy 23–34	**189 ± 79 ms**	—	1.5T FAIR, PLDs 400–2600 ms
Shirvani 2019, n=12 healthy	290.77 ± 70.07 ms (BAT)	184.21 ± 37.93 ms (BAT)	3T FAIR 3D-GRASE, 14 TIs
Harteveld 2020, n=15 healthy 51 ± 10	**0.47 ± 0.13 s**	0.70 ± 0.10 s	3T multi-delay **FAIR**
Harteveld 2020, same subjects	**0.71 ± 0.25 s**	0.86 ± 0.12 s	3T multi-delay **pCASL**
Garcia-Ruiz 2025, n=16 healthy	female 0.75 ± 0.11 s / male 0.96 ± 0.22 s (1.5T, pooled)	—	multi-delay PCASL
Shimizu 2017, n=14 males	younger 961.33 ± 260.87 ms; older 1227.94 ± 226.51 ms	—	3.0T pcASL
Kim 2017, 46 kidneys of 23	1141 ± 262 ms	1123 ± 245 ms	3T pCASL *(abstract only)*
Ahn 2020, 7 native / 4 transplant	native 641 ± 141 / **transplant 919 ± 49 ms**	746 ± 150 / 935 ± 81 ms	3.0T pCASL *(abstract only)*

Note two orderings that **contradict each other**: Harteveld and Ahn find medulla ATT > cortex ATT (physiologically expected, since medullary blood traverses cortex first); Shirvani finds the reverse. Kim finds cortex ≈ medulla, i.e. no separation at all.

> ⚠️ **Do not build an ATT threshold.** Kim 2017's own conclusion is that "*ATT for the renal cortex and medulla has poor reproducibility and high variation*" — its medullary and ATT ICCs were poor, and Harteveld reports an ATT ICC of 0.32 with CV 33.6%. A parameter that does not reproduce within a subject cannot anchor a PASS/FAIL rule.
>
> ⚠️ **And do not treat the transplant ATT difference as established.** Ahn 2020 is **n=4 vs n=7** with a **17-year age gap** running in the direction that *depresses* cortical perfusion — yet the transplants measured *higher* RBF (358.3 vs 271.8). The iliac-fossa geometry explanation is plausible physiology; this dataset does not establish it.

The one thing ATT is genuinely useful for right now: checking that the chosen PLD clears it. Renal cortical ATT sits roughly 0.19–1.14 s across all these studies, and the consensus PLD of 1.2–1.5 s comfortably exceeds it — which is *why* single-delay is defensible in the kidney.

A useful implementation default: Bones 2022's automated pipeline assumed **ATT = 750 ms** for single-PLD pCASL quantification. Tag it `implementation`; it matches no measured value in the table.

5.6 Renal T1 — you need this for the M0 correction

The consensus mandates `SI_PD / (1 − exp(−TR/T1_tissue))` and supplies no T1. From Wolf et al. 2018 (systematic review, healthy subjects):

Wolf 2018 — renal T1 values by field strength

FIELD	CORTEX T1	MEDULLA T1
1.5T	~827 – 1080 ms	~1054 – 1428 ms
3T	~1124 – 1406 ms	~1388 – 1685 ms

The most-cited single reference is **de Bazelaire et al. 2004** (*Radiology* 230(3):652–9) — cortex **1142 ± 154 ms**, medulla **1545 ± 142 ms** at 3.0T (n=6); cortex 996 ± 58, medulla 1412 ± 58 at 1.5T (n=4). *(Paywalled — I have read these only via the Wolf 2018 table, so treat as secondary.)* Note David Alsop is a co-author of de Bazelaire too.

Directly measured alongside PCASL by Garcia-Ruiz 2025 (n=16): cortical T1 **1023.29 ± 39.30 ms** (1.5T) / **1356.02 ± 38.72 ms** (3T); medullary **1348.54 ± 50.40** / **1632.68 ± 46.24 ms**.

> ⚠️ **Two traps.** (1) The **cortico-medullary T1 gap is 300–450 ms**, so a single "kidney T1" in the M0 correction is itself an approximation — evaluate per compartment if you can. (2) Stanisz 2005 is widely cited for tissue T1 but its "kidney" row is **rat kidney measured in vitro** (1194 ± 27 ms at 3T, 690 ± 30 ms at 1.5T). The 1.5T value is ~140 ms below any human in vivo cortex T1. **Do not use it as a human renal constant.**

5.7 Repeatability — the ceiling on any renal threshold

Odudu 2018 Table 1 — reproducibility by study, cortex vs medulla

Pooled ranges (Odudu 2018, from the 17 of 53 studies reporting reproducibility):

ROI	INTRA-VISIT	INTER-VISIT
Cortex	ICC 0.62–0.98, CV **3–18%**	ICC 0.85–0.97, CV **4–13%**
Medulla	ICC 0.27–0.94, CV **3–43%**	ICC 0.13–0.96, CV **4–37%**

> ⚠️ **The review contradicts its own table.** Those ranges come from the narrative text. Table 1's lowest medulla intra-visit ICC is **0.42**, not 0.27, and its highest cortex intra-visit CV is **20%**, not 18%. Cite the narrative sentence and be ready for a reviewer who opens the table.

Individual studies:

STUDY	DESIGN	CORTEX	MEDULLA
Hammon 2016	1.5T FAIR True-FISP, 3 × 10-min per occasion, 14 d apart	ICC 0.97, CV 4.19 ± 1.33%	ICC 0.96, CV 4.12 ± 1.36%
Gillis 2014	3.0T FAIR True-FISP, 4–28 d apart, n=12	ICC 0.85 (0.65–0.94), CVws **9.2%**	not measured
Echeverria-Chasco 2023	3.0T pCASL, n=18 allografts	CVws **10.73%**, ICC 0.84	CVws 16.41%, ICC 0.61
Garcia-Ruiz 2025	1.5T+3T PCASL, n=14, ~1 wk	wsCV 7.27–11.33%, ICC 0.72–0.83	wsCV 10.01–19.38%, ICC 0.08–0.43
Olsen 2025	3T FAIR + PET, n=10	same-day CoV 10.9%; between-day (median 16 d) **11.2%**	not measured
de Boer 2021	3T FAIR, n=13	CoV 10%, ICC 0.47 (0.13–0.72), **repeatability coefficient 29% (19–39%)**	not reported
Harteveld 2020	3T, n=12 at visit 2, 1 wk	**FAIR 9.9%** / **pCASL 33.9%**	FAIR 13.8% / pCASL 30.9%
Buchanan 2018	3T FAIR, n=10, 2 visits ≤2 wk	between-session CoV **SE-EPI 17.2%** / **GE-EPI 28.3%**	—
Nery 2019	1.5T FAIR, **n=11 children**, 23 ± 10 d	inter-session WSCV **20.1%** with motion correction, **50.6%** without	—
Cutajar 2012	1.5T FAIR, **intra-session**	SDws 14.6 mL/min/100 g, **CVws 7.52%**	— *(abstract only)*

> 🎯 **The numbers that actually constrain your design:**
>
> - Best-case **intra-session** cortex ≈ **7.5%** (Cutajar). Realistic **inter-session** cortex **~10–11%** (Garcia-Ruiz, Olsen, de Boer, Echeverria-Chasco all converge here).
> - **Inter-observer** ROI drawing is only **1.30–2.46%** (cortex, Garcia-Ruiz; ICC 0.96–0.99), and Radovic 2022 measured interobserver ICC **0.96 (0.92–0.98)**. **So segmentation is NOT the dominant error — acquisition and physiology are.** Put your QC effort into Stream A motion, M0 and labelling checks, not ROI refinement.
> - de Boer's **repeatability coefficient of 29%** is the honest answer to "how big must a change be before it is real?"
>
> ⛔ **But note the category error waiting here.** All of these are **wsCV / ICC / repeatability coefficients** — group statistics computed across subjects × sessions. They **cannot be evaluated on a single incoming scan**, which is what a per-scan PASS/FAIL check must do. They bound a *longitudinal-change* alarm; they are not per-scan thresholds. And Gillis's 9.2% is a **between-session** wsCV, **not** a spatial coefficient of variation across voxels — if anyone calibrates a renal sCoV threshold at 9.2%, essentially every real kidney map will fail.

5.8 Image-quality metrics — the closest thing to renal QC reference values

This is the single most useful table in the document for Stream B, and it comes from **Garcia-Ruiz et al. 2025** (n=16 healthy, 8 female, 41.8 ± 13.3 y, eGFR 101.40 ± 13.72; **same subjects at both field strengths on the same day**; PCASL, SE-EPI, 3 slices per kidney, 3×3×5 mm, TR 6000 ms, PLD 1.3 s single-delay, 2 background-suppression pulses):

Garcia-Ruiz 2025 Table 3 — PWS, spatial SNR, tSNR, BS efficiency

METRIC	CORTEX 1.5T	CORTEX 3T	MEDULLA 1.5T	MEDULLA 3T
PWS (% of M0)	2.95 ± 0.56	3.09 ± 0.59	1.36 ± 0.15	1.40 ± 0.19
Spatial SNR	1.10 ± 0.35	2.05 ± 0.66	0.54 ± 0.12	0.92 ± 0.26
tSNR	3.54 ± 0.71	3.33 ± 0.54	2.00 ± 0.44	1.89 ± 0.41
BS efficiency (%)	91.34 ± 3.18	88.45 ± 3.02	91.81 ± 4.02	86.40 ± 4.52

Their spatial-SNR definition is implementable and renal-specific:

```
Spatial SNR = 0.655 · μ_signal / ( sqrt(N_pairs) · σ_noise )
```

where `μ_signal` is the mean in a kidney ROI and — the clever part — `σ_noise` **is measured in a LIVER ROI**, *"since blood flow to the liver was not labeled; therefore, the difference signal in the liver should be noise."*

> ⛔ **Get the square root right.** It is `sqrt(N_pairs)` in the denominator, not `N_pairs`. At 20 pairs that is a factor of **4.5** difference, and getting it wrong makes your values incomparable to the reference table above. (This is exactly the kind of thing that gets lost when a paper's MathML is converted to plain text.)

Readout dependence — Buchanan, Cox & Francis 2018 (n=10 healthy, 27 ± 10 y, 5 female, eGFR>60, 3T Philips Achieva, FAIR + FOCI, respiratory-triggered, 25 pairs, renal **cortex**):

Buchanan 2018 — image quality by readout scheme

READOUT	PWI-SNR	TSNR	VAR ΔM ACROSS SLICES	CORTEX RBF
bFFE	6.2 ± 3.6	2.4 ± 2.0	26 ± 11%	276 ± 29
GE-EPI	6.3 ± 1.5	1.5 ± 0.8	20 ± 5%	222 ± 18
SE-EPI	4.9 ± 1.5	**2.6 ± 1.6**	**11 ± 3%**	201 ± 36
TSE	8.5 ± 4.1	2.4 ± 1.8	20 ± 4%	200 ± 20

> ⛔ **The most dangerous number in this document.** Buchanan defines tSNR as "*the mean PW signal divided by the standard deviation across the 25 ASL pairs*" — a **single-repetition** tSNR. The **averaged** perfusion map has roughly √25 = **5×** higher SNR. Anyone who sets an osipy-qc renal SNR threshold from "tSNR 1.5–2.6" would set it about 5× too strict and fail nearly every good kidney scan.
>
> ⚠️ Note also `var ΔM` is defined as "*the standard deviation in the PW signal across the slices divided by the mean PW signal*" — i.e. it is a **coefficient of variation across slices**, despite the name. Do not implement it as a variance.

Cross-scheme tSNR (Franklin 2021, n=5 analysed of 6, ages 23–30, 3T, renal cortex): VSASL 1.59 ± 0.21, AccASL 1.54 ± 0.41, mm-VSASL 1.37 ± 0.34, VSI-ASL 1.62 ± 0.36, pCASL 1.79 ± 0.56, **FAIR 4.61 ± 0.71**. Medulla VSI-ASL was **0.17 ± 0.14** — effectively zero. Independently, Bones 2021 found FAIR cortical tSNR 3.30 ± 0.72 vs VSASL 0.82–1.37 at 1.5T.

Background suppression roughly doubles tSNR (Bones 2019, n=9, 1.5T whole kidney): 0.60 ± 0.15 without BGS → 0.93 ± 0.22 with heavy BGS2H (ASL-driven registration), a ~1.6× gain; the fat-driven registration arm went 0.44 → 0.86, ~1.95×.

> 🚩 **But BS and registration trade off against each other.** Same paper: with heavy background suppression there is so little anatomical contrast left that conventional ASL-image-driven coregistration to M0 **succeeded only 54% of the time**, versus 100% when driven by separately acquired fat images. **A renal QC tool must check registration quality separately from BS quality.**

> ⛔ **These tSNR values span roughly 0.17 to 4.61 across the four sources above**, with incompatible definitions (single-pair vs averaged; whole-kidney vs cortex; 1.5T vs 3T; six labelling schemes). tSNR is only comparable within a fixed (definition, ROI, readout, labelling, field-strength) tuple. **There is no single renal tSNR threshold to be had.**

5.9 Two special populations: transplants and children

Transplants — a real band exists, but it discriminates *function*, not *quality*

COHORT	CORTICAL RBF (ML/100 G/MIN)	N	FIELD	SOURCE
Stable allograft	196.54 (53.22) → 188.35 (49.44)	18 analysed, 54 ± 14 y, 120 ± 105 mo post-Tx	3.0T pCASL	Echeverria-Chasco 2023
eGFR ≥60	259.74 ± 47.52	29 of 58, 35.4 ± 12.5 y	1.5T pCASL 3D-GRASE	Que 2026
30 ≤ eGFR <60	166.50 ± 19.79	18	"	"
eGFR <30	112.76 ± 32.08	11	"	"
Stable (control)	277.69 ± 67.17	20	1.5T	Ma 2025 ⚠️
Chronic allograft dysfunction, grade 3	92.89 ± 35.62	19	"	"
Normal graft	191.84 ± 63.96	42, 35.69 ± 10.06 y	3.0T 3D pCASL	Peng 2023
Injured graft	104.33 ± 54.76	93, 39.89 ± 11.26 y	"	"
Long-term survivor, normal–mild	221.2 ± 52.1	30, 43.4 ± 9.1 y	1.5T pCASL	Jiang 2024
Long-term survivor, severe	122.6 ± 46.5	13	"	"

Published discrimination cut-offs: **207.53** (AUC 0.967, Que 2026), **210.06** (AUC 0.90, Ma 2025), **192.4** (AUC 0.739, Zhang 2025), **88** (AUC 0.729, Jiang 2024). A meta-analysis of 28 studies / 1,956 recipients (BMC Nephrology 2026) gives a mean difference of **98.48 mL·min^{-1}·100 g^{-1}** (95% CI 68.66–128.31) between eGFR ≥60 and <60, with **I^2 = 57%**.

> ⛔ **Those cut-offs answer a CLINICAL question — "is this allograft working?" — not a QC question. Never reuse them as image-quality thresholds.** They are also not as independent as they look: Que 2026 and Jiang 2024 share identical acquisition parameters (1.5T, pCASL, 3D GRASE, TR 3963 ms, TE 15 ms) and identical eGFR strata, yet sit ~40 units apart at nominally equivalent function.
>
> ⚠️ **Ma 2025 has an internal unit inconsistency**: its Table 1 header reads `RBF (mL/100 g/min)` while the Results text reads `mL/min/1.73 m²` for the same values. Treat its absolute numbers with caution.
>
> ⚠️ **Direction is not established.** The headline "transplants run lower than native" does not survive: Ahn 2020 measured transplant cortex **higher** (358.3 vs 271.8), and pCASL native values of 197–201 sit below most stable-allograft values. Ship a transplant **profile** (geometry, labelling-plane position, gating expectations — all well evidenced) **without** an absolute band.

Children — the honest answer is that normative data do not exist

I found **four** paediatric renal ASL papers in total, covering roughly **50 children**. **None** is in healthy children. **None** is in neonates. **None** reports a cortex/medulla split.

STUDY	COHORT	VALUE	PROTOCOL
Nery 2019	**11 children, 12 ± 3 y (7–17), severe CKD**, eGFR 26 ± 9	RBF **95.3 ± 46.9** (registration) / 94.7 ± 47.3 / 127.4 ± 81.8 (no motion correction) — ROI is "*cortical and/or functional renal parenchyma*"	1.5T Avanto, FAIR Q2TIPS 3D-GRASE, 2 BS pulses, TI 1200 ms, TI1 900 ms, respiratory-triggered, **25 pairs**
Harteveld 2022	**10 children, mean 4.3 y (1.7–8.3)**, abdominal tumours; most under general anaesthesia	**No absolute perfusion.** PWS only: 1.5 ± 0.61% @ PLD 0.5 s → 0.51 ± 0.27% @ 1.5 s in the contralateral *normal* kidney	1.5T pCASL, LD 1.5 s, coronal 2D GE-EPI
Radovic 2022	**18 paediatric/young-adult transplant recipients**, median 16 y (range **5–29**)	Good function 219.89 ± 57.24; impaired 146.22 ± 41.84 (p=0.0066); overall 85–303, mean 183.06 ± 61.66. AUC 0.864	1.5T FAIR True-FISP + 3D-GRASE + BS, TI 2000 ms, free breathing
Radovic 2025	10 paediatric recipients 14 ± 4 y + **10 adult donors 46 ± 6 y**	Recipient per-kidney: 251.71 ± 78.27 → 486.61 ± 156.08 at 3 mo (p=0.0001). **Units are ml/kidney/min, not per 100 g**	1.5T FAIR 3D-GRASE, TI 2000 ms, **6 control/tag pairs**

Nery 2019 — paediatric CKD repeatability with and without motion correction

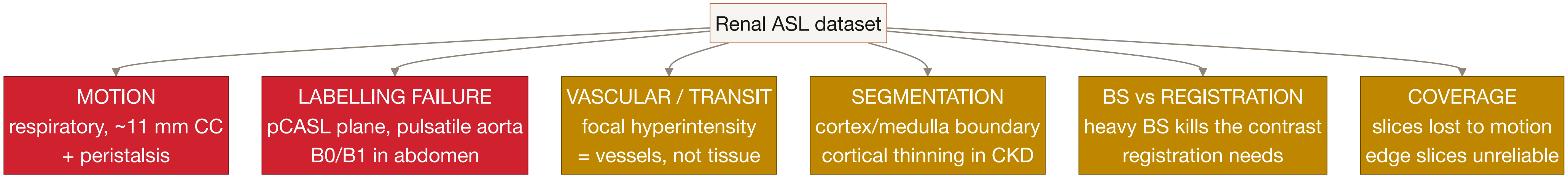
What the paediatric literature *does* give you, and it is genuinely useful:

- **A realistic failure rate.** Radovic 2022: "*Three patients were excluded due to poor cooperation and motion artefacts*" — 3 of 21 = **14.3%** (95% CI roughly 3–36%, so treat it as "order of ~10%", not a calibrated target).
- **A measured motion-correction benefit.** Nery 2019: inter-session WSCV **0.506 → 0.201** and ICC **0.372 → 0.833** when retrospective registration is applied.
- **Age-scaled sensitivity.** In the same paper the tSNR improvement from motion correction correlates negatively with age (R = –0.37, P = 0.015 for pipeline A vs B; R = –0.49, P = 0.0009 for A vs C), "*showing that these MC methods are especially useful in younger children.*" ⚠️ *A separate correlation in that paper, R = –0.38 / P = 0.014, is between age and **Euclidean distance** — a motion index — not entropy. Do not mislabel it.*
- **A published warning about using repeatability as a quality proxy** — see §6.6.
- **A practical constraint** (De Mul 2025 review): "*sedation or general anesthesia with IV access may be necessary for children under 7 years old or those who are uncooperative*" — and anaesthesia itself alters perfusion. Encode as context, not failure.

> 📌 **The citable authority for shipping paediatric renal thresholds** `uncalibrated` is De Mul et al. 2025 (*Pediatr Nephrol*): "*Arterial spin labeling (ASL) is an experimental and not validated technique that enables the measurement of cortical and medullary perfusion without the need for exogenous contrast agents.*"

> ⚠️ **A protocol tension worth knowing.** Radovic 2025 used only **6** control/tag pairs against the consensus's ≥20, and says why: "*increasing it would extend the acquisition time to 12 min and 6 s, which was unacceptable in our (or any) study group.*" But this does **not** mean a ≥20 check would fail every paediatric study — Nery 2019 used **25** pairs. And R4.7's body text scopes the rule to "*when using the recommended 2D readout*", while both paediatric studies used **3D-GRASE**. **Gate the pair-count check on readout type.**

PART 6 — The failure modes, and what each looks like in the data


Renal ASL dataset
MOTION
respiratory, ~11 mm CC
+ peristalsis
LABELLING FAILURE
pCASL plane, pulsatile aorta
B0/B1 in abdomen
VASCULAR / TRANSIT
focal hyperintensity
= vessels, not tissue
SEGMENTATION
cortex/medulla boundary
cortical thinning in CKD
BS vs REGISTRATION
heavy BS kills the contrast
registration needs
COVERAGE
slices lost to motion
edge slices unreliable

6.1 Respiratory motion — the dominant mode

Signature: Bones 2019 — "*When motion in free-breathing renal ASL series is not accounted for, perfusion maps have a blurred appearance with low SNR due to subtraction artifacts.*"

Cost when uncorrected: Nery 2019's paediatric inter-session WSCV was **0.506 without motion correction** versus **0.201 with it** — a 2.5× degradation.

Whole-subject loss is real: Bones 2019 lost **1 of 10** subjects to through-plane motion — "*One subject indicated difficulties with the paced-breathing protocol, resulting in major kidney displacements with in-plane and through-plane motion corrupting the PWIs.*" Franklin 2021 lost **1 of 6** for the same reason.

Why coronal-oblique matters here: it puts most respiratory displacement **in-plane**, where 2D registration can fix it. Through-plane motion is what registration cannot rescue — which is why an orientation check is a *motion-risk* check.

> 🚩 **An external respiratory trace is not sufficient.** Bones 2020 found "*no one-to-one correlation between respiratory bellows signal change and generated spurious label. Some repetitions had tissue labeling without the bellows indicating motion, others showed no spurious labeling even when motion was detected.*" **Image-derived metrics are required.**

Published motion-handling rules you can implement:

RULE	VALUE	SOURCE	TIER
Reject subtraction if **>20% of kidney voxels beyond ±2 SD** of the voxel-wise mean	—	Harteveld 2020, n=15, 3T. Fired in **18/27 FAIR** and **19/27 pCASL** datasets, max 2 pairs excluded per delay	`implementation`
Keep ΔM only if **>80% of kidney voxels within ±2 SD**	authors call it "*an empirically chosen threshold*"	Bones 2021, n=7, 1.5T	`implementation`
Reject subtraction if **>20% of voxels beyond ±1.5 SD**	~1 of 10 pairs rejected (range 0–3)	Harteveld 2022, **n=10 children**, 1.5T pCASL	`implementation`
Discard time-series outliers beyond **mean ± 2 SD**, plus per-slice fit **RMSE > mean + 2 SD**	**1% of slices excluded**; no participant lost entirely	Garcia-Ruiz 2025, n=16	`implementation`
Discard volume if **cortical ROI signal outside mean ± 2 SD**	scalar-per-volume, cheaper	Echeverria-Chasco 2023, n=18 allografts	`implementation`
Exclude difference image if kidney movement **>1 voxel**	visual inspection, voxel size unstated	Cox 2017, n=127	`implementation`, under-specified

> 💡 **Note that Harteveld's rule fires in roughly two-thirds of NORMAL healthy datasets.** "The rule fired at all" is therefore not a defect signal — only the *count* matters, and the observed ceiling in good data is 2 pairs per delay.

Navigator gating (Tan et al. 2014, n=10 healthy 42 ± 16 y + 5 CKD, 3T FAIR True-FISP — ⚠️ *PDF not in my local library; figures read from the PMC record*): an **8 mm acceptance window** (2 navigator pixels) left residual displacement of **max 4 mm SI, average 1.1 mm SI, <1 mm medial-lateral**, at an acquisition efficiency of **50 ± 13%** (coronal navigator) / 41 ± 9% (sagittal). SNR was 36.65 / 31.36 with navigators versus **12.15 for breath-hold** — quantitative support for the anti-breath-hold recommendation.

> ⚠️ **Do not turn 41–50% acquisition efficiency into a threshold.** Those *are* the normal healthy means; a "below 40–50% is poor compliance" rule would fail about half of normal sagittal-navigator scans. Report it as INFO.

6.2 Labelling failure

The renal-specific causes have no brain analogue. From the Taso renal ASL symposium slides (⚠️ *conference slides, not peer-reviewed*): "*Off-resonance/B1 effects: Dielectric (B1) and field inhomogeneities (B0) are more important in the abdomen. Impacts labeling efficiency + imaging quality. Highly pulsatile flow: The abdominal aorta has a strong pulsatile profile. Affects labeling efficiency (especially in PCASL).*"

Taso — renal-specific ASL challenges

A published quantitative signature: Harteveld 2020 — "*For some pCASL data sets (3 out of 27), the averaged perfusion-weighted images showed very low signal corresponding to cortical RBF values < 100 mL/min/100 g. Compared with RBF values obtained at the other visit or with FAIR labeling in the same subject, it seems that these pCASL measurements failed.*"

> ⚠️ **But you cannot ship `<100 = FAIL`.** (a) The authors identified failure by comparing against the *same subject's other visit and other labelling scheme* — references you will not have at QC time. (b) It is pCASL-specific, from a >40 y cohort, on a gradient-echo readout the consensus does not recommend. (c) Genuinely diseased kidneys live below it: allografts with biopsy-proven glomerulonephritis have cortical ASL median **65.38** (IQR 43.14–84.11) mL/min per 100 g, and impaired-function allografts sit at 146.22 ± 41.84. **A low-CBF flag must be structural — a spatial pattern — not a bare number.**

FAIR has its own failure mode with a diagnostic rule (preclinical, Chuang 2021): "*If perfusion signal is weak, the nonselective inversion may not be effective. Try increasing the bandwidth of the inversion pulse.*" And "*Shimming is particularly important for nonselective (global) inversion, since adiabatic condition depends on B0 field homogeneity. Shimming should be performed on a global level.*"

Velocity-selective ASL has a beautifully automatable failure signature. Bones 2020, n=15 healthy: spurious labelling from motion appears as "*homogeneously high ΔM over the entire kidney ROI*" — i.e. an abnormally high whole-kidney mean **with loss of cortico-medullary contrast**. It affected **25.4% of repetitions** with feet-head labelling (4.3% RL, 4.9% AP), and **5 of 15 subjects were excluded**. That two-part signature — high mean *and* flat contrast — is directly implementable and is the best concrete design lead I found.

6.3 Vascular / transit contamination

The consensus's remedy is manual:

> *"ROIs should be adjusted to avoid hyperintense signals on the perfusion image as they likely represent vessels."*

Gillis 2014 implemented it as a percentile clip: "*Pixels with intensity at the extremes of the range were excluded, as these were likely to represent adventitia or major vessels.*"

Olsen 2025 used a hard voxel cap: "*Segmentation of the renal cortex was performed on the averaged perfusion-weighted images and used to calculate global perfusion values, **excluding voxels exceeding 500 mL/min/100 mL**.*"

> 📌 **Trace that 500 properly before you use it.** Olsen cites it to two references; one is the Nery consensus, and I searched the consensus — `500` appears exactly twice, both times as "500 μs" RF pulse duration. **There is no perfusion cap in the consensus.** The real origin is **Gullaksen et al., Diabetologia 2023;66:813–825**: "*Perfusion values outside 0–500 ml (100 g)$^{-1}$ min^{-1} were rejected.*" Three consequences: (1) cite Gullaksen, not PARENCHIMA; (2) the origin is **per 100 GRAMS**, Olsen re-states it per 100 mL; (3) the original rule is **two-sided** — it also rejects negative perfusion, which is the direct renal analogue of your brain negative-voxel check, and is the more useful half.

A 2026 deep-learning method (Magn Reson Imaging 126:110573, paywalled — abstract only) names "*low signal-to-noise ratio, large vessel contamination, and the absence of magnetization direction information in magnitude images*" as the three challenges of multi-TI renal FAIR, and builds an automatic large-vessel exclusion mask. Its own abstract cautions that in vivo differences between methods "*cannot be explained by their differences observed in simulation*" — so do not lean on it.

> ⛔ **There is no renal spatial-CoV precedent.** A Europe PMC search for spatial coefficient of variation crossed with renal/kidney returns hits that are, without exception, cerebral. And a naive whole-kidney sCoV would be dominated by the **normal 2.3–7× cortico-medullary gradient**, not by artefact. If you build one, it is novel and `uncalibrated`.

6.4 Segmentation failure

Signature: cortex:medulla ratio approaching 1 (§2.2), and/or fitted tissue T1 values that collapse together.

A disease-driven version with no brain analogue — the consensus:

> *"In the case of advanced disease, the reduction in cortical thickness can be severe enough such that it approaches the typical dimensions of the ASL voxels, significantly reducing the number of pure cortex voxels from which cortical RBF can be estimated. The mixing of perfusion signal from the cortex with medullary signal may bias cortical RBF estimates to lower values (i.e. potentially causing an apparent reduction of cortical perfusion). … Therefore, cortical RBF results from ASL should be interpreted with caution in cases where cortical thinning is evident."*

In advanced CKD, partial volume can *manufacture* a low cortical perfusion reading. A cortical-thickness-to-voxel-size ratio would be a good flag — but it is an idea, not a published check, and there is no numeric trigger.

6.5 Background suppression vs registration — the interaction

Covered in §5.8, but it deserves its own line because it is a genuine design constraint: heavy BS improves tSNR ~1.6–2× **and simultaneously destroys the anatomical contrast registration depends on**, dropping ASL-driven M0 coregistration success from 100% to **54%** (Bones 2019, n=9, under the heaviest BGS4H condition). **Check BS quality and registration quality separately.**

Also note the consensus's BS bookkeeping: "*The efficiency, α, should be multiplied by an additional factor of 0.93 for each BGS inversion pulse added [R9.9]*", with 2–4 pulses typical. And that magnitude-only subtraction leaves **positive** residual static tissue signal (Bones 2019) — so a residual-static-signal check must not assume perfect cancellation.

6.6 The repeatability trap — read this before designing any QC metric

Nery 2019 documented a failure mode that would have caught this project out:

> "*Despite the prevalence of artefacts in the RBF maps without applying MC methods, the intra-session repeatability was high. Because both ASL runs shared the same M0 (obtained from the same PD acquisition) systematic errors because of a consistent misalignment between the M0 and the ASL data set in both ASL runs resulted in a repeatable mean RBF despite severe corruption in the RBF maps in the no-registration case.* ***For this reason, intra-session WSCV and ICC alone should not be taken to be indicators of image quality directly.***"

Two scans can agree perfectly and both be wrong, because they share a broken denominator.

6.7 Coverage loss

Measured (Olsen 2025, 60 single-kidney scans, 5-slice protocol, 3T FAIR): only **48.3%** of scans retained all five slices. 25.0% used four, 16.7% three, 8.3% two, and 1.7% were reduced to a single slice. Slices were dropped for "*motion compensation failure*" and localised artefact — by **visual assessment**, which is itself the gap.

Edge slices are systematically worse (Radovic 2022): "*maps with significant noise were excluded (mostly generated from slices at the very beginning or end of the imaging stack, mainly due to partial volume effect with the nearby structures). The middle slice … was the most representative one.*"

> ✅ **Design consequence:** a **per-slice** verdict with position weighting, plus a "usable slice fraction" INFO metric. But 48.3% is one protocol on one scanner in healthy young adults — it is motivation, not calibration.

6.8 The state of the art in renal ASL QC, as of now

> "*Data exclusion was performed based on visual assessment of the averaged perfusion-weighted images. Whole slices were excluded from the segmentation in cases of motion compensation failure. Smaller image regions suffering from artifacts were also excluded from the segmentation.*" — **Olsen et al. 2025**, a state-of-the-art simultaneous ASL/[^{15}O]H_2O PET study

> "*Quality of all data with regard to image artifacts was visually assessed.*" — **Bones et al. 2022**, the most automated renal ASL workflow published

Even the preclinical protocol chapter specifies QC only qualitatively: "*Scans with poor quality or large movement should be discarded*", checking for "*sudden movement, spikes, banding artifacts, or sudden changes in SNR.*"

Renal ASL quality control in 2026 is still done by eye. That is the gap.

PART 7 — What is NOT known

> 💎 **This section is as valuable as the rest of the document.** Stating the gaps plainly is what makes the numbers you *do* ship believable. Every item here is a verified absence, not a "I didn't look hard enough".

7.1 There is no renal QEI, and no renal quality index of any kind

No composite quality metric for renal ASL exists. No paper defines one, and the consensus contains no numeric image-quality threshold (§4.6).

Why QEI cannot simply be ported:

QEI COMPONENT	BRAIN INPUT	KIDNEY EQUIVALENT
Structural similarity: `Pearson(CBF, 2.5·GM + 1·WM)`	GM/WM probability maps, free with any T1w	❌ **none exist.** Consensus default is manual ROI (R9.10)
Dispersion: pooled variance / mean GM CBF, masks at prob ≥ 0.9	three tissue classes with probabilities	❌ two compartments, binary masks, boundary Dice ≈ 0.78
Negative GM fraction	GM mask	⚠️ possible on a cortex mask — this is the one term that ports
Calibration	fitted to 2 neuroradiologists rating 101 maps; cutoff ≈ 0.5	❌ no rated renal dataset exists

The partial analogue that *does* exist: Renal Segmentor (and UKAT, which wraps it) can emit a per-voxel **whole-kidney probability map** rather than a binary mask. That supports a whole-kidney structural-correlation term. It does **not** support a two-compartment one.

The nearest thing to a defined renal quality panel is Buchanan 2018's three metrics — PWI-SNR, tSNR, and inter-slice variance of the PW signal — defined explicitly and with **no thresholds attached to any of them**. That is exactly the gap.

7.2 There is no renal reference perfusion value that survives protocol

Established in §5.2: healthy adult cortex spans **139–441 mL/100 g/min** across published means, driven by labelling scheme (~80%), field strength (~11%), age (~20%) and ROI definition (~7%). The healthy and patient ranges overlap almost entirely.

Even the **gold standard disagrees with itself**. Olsen 2025 compared ASL against simultaneous [^{15}O]H_2O PET in the same 10 healthy subjects:

Olsen 2025 — ASL vs PET agreement

- ASL **150–422** vs PET **184–470** mL/min/100 mL across individuals
- Method-comparison bias **18**, **limits of agreement ±136** mL/min/100 mL

- *"Agreement between ASL-MR and PET was acceptable at perfusion values between approximately 250 and 350 mL/min/100 mL"* — i.e. only over a narrow middle band
- Repeatability CoV: ASL 10.9%, PET 8.4%; reproducibility (median 16 days apart) ASL 11.2%

> ⚠️ **Do not confuse the two LoA figures.** ±136 is the **method-comparison** limit. The narrower −56 to +67 (same-day) and −98 to +100 (between-day) are **repeatability** limits. Quoting the repeatability figure under an agreement claim makes ASL–PET concordance look about twice as tight as it is.

7.3 No kidney tissue T1 in the consensus

The consensus mandates the M0 incomplete-relaxation correction and supplies no `T1_tissue`. You must source it externally (§5.6), it differs by ~300–450 ms between cortex and medulla, and the published healthy ranges are themselves 250 ms wide at each field strength.

7.4 No inter-centre reproducibility study

Odudu 2018 states it plainly:

> *"At the time of this review, there are no published studies comparing the reproducibility of ASL at different magnetic field strengths or under different labelling approaches. Similarly, we found no studies of reproducibility between centres."*

Two of the three are now **superseded**: Garcia-Ruiz 2025 is a within-centre 1.5T-vs-3T comparison, and Harteveld 2020 is a within-centre FAIR-vs-pCASL comparison. **The inter-centre gap still stands** as of my search — the only multi-centre, multi-vendor renal PCASL reproducibility work I could find is an **unpublished ISMRM 2026 abstract** (Echeverria-Chasco, "*Reproducibility of renal PCASL sequences at 3.0 T: preliminary results from a multicenter and multivendor study*"). A conference abstract is not a citable number.

> ⚠️ Date-bound the Odudu quote to its **January 2018** literature cut-off. Do not restate it in the present tense.

7.5 No outlier-rejection method or threshold, despite unanimity

R8.2 is 0% abstention / **100% agreement** that outlier rejection is recommended — and specifies neither a method nor a number. The consensus survey of practice is itself an admission:

> *"Outlier rejection methods, including retrospective sorting of renal ASL data, have relied on manual or automatic approaches, including using data from external sensors such as respiratory bellows."*

The five implementation-tier rules in §6.1 are individual groups' choices, not validated cut-offs.

7.6 BIDS does not cover the kidney

- The ASL-BIDS authors say so: "*Whereas ASL-BIDS could perhaps be used for other body parts, **ASL-BIDS is validated in ASL images of the brain only***" — and cite the renal consensus as the out-of-scope example (Clement et al. 2022, *Sci Data* 9:543).
- The words *kidney*, *renal* and *abdominal* appear **zero times** in the released BIDS specification schema (v1.11.1, schema 1.2.1). *brain* appears 64 times.
- The only quantified-perfusion volume type BIDS defines is **`cbf`** — "*The **cerebral** blood flow (CBF) image … quantified into mL/100g/min*", defined by reference to the 2015 brain White Paper. **There is no renal-blood-flow volume type.** The raw types (`control`, `label`, `m0scan`, `deltam`) are organ-neutral and port fine; the derivative type does not.
- `M0Estimate` is defined as "*A single numerical **whole-brain** M0 value*".
- The organ hooks that do exist are **`BodyPart`, `BodyPartDetails` and `BodyPartDetailsOntology`** — all **OPTIONAL**, free-text, on the MRI sidecar rule. There is no filename entity for organ on `*_asl`; the `voi-<label>` entity exists only for MRS.
- **Good news:** respiratory traces *are* legal alongside renal ASL — the physio file rule permits `*_physio.tsv.gz` in the `perf` datatype, and `respiratory` is a defined column. So `perf/sub-01_recording-respiratory_physio.tsv.gz` is spec-compliant. The breathing **signal** has a standards-compliant home; the breathing **strategy** does not.
- Roughly **eight** Nery Table 4 items have no BIDS field: physiological triggering/gating, fat suppression, pixel bandwidth, slice gap, field of view, number of slices, image orientation, quantification model. *(Caveat: `ScanOptions`, which maps to DICOM (0018,0022), is where a DICOM-faithful converter puts fat-sat and respiratory gating — so "no **dedicated typed** field" is the precise statement. And orientation, FOV and slice count are recoverable from the NIfTI affine.)*

> 🐛 **A genuine, apparently unreported BIDS bug that lands on the renal FAIR path.** `LabelingSlabThickness`'s description says "*For non-selective FAIR a zero is entered*" — but the schema constrains it to `exclusiveMinimum: 0`. A non-selective FAIR acquisition therefore cannot be encoded as the prose instructs without failing validation. I found **zero** BIDS issues mentioning the field. Since FAIR is the dominant renal PASL variant, this is a cheap, high-visibility first upstream contribution. *(Severity note: the field is only `recommended` for PASL, so the workaround is to omit it.)*

Also worth knowing, because a BIDS-literate mentor will: there is no *registered, numbered* BEP for body/abdominal imaging — but bids-specification issue **#1569, "BIDS-BodyPart extracted from BEP025", is OPEN**, filed by the BEP025 lead, proposing to extend body-part tagging "*to other parts of the body*". So "nobody has proposed **renal** BIDS" is true; "nobody has proposed **organ** BIDS" is not.

7.7 There is no renal ASL processing pipeline

- **ASLPrep, ExploreASL, PyASL, BASIL/oxford_asl** contain **zero** kidney or renal code. I grepped the local clones: `kidney` = 0 files across all of them; the single `renal` hit in ASLPrep is a false positive inside a base64-encoded image blob in `docs/_static/brainextraction.svg`.
- The OSIPI TF2.2 ASL code collection has **0** renal matches across 414 tracked paths; its only non-human content is **mouse brain** ASL.

- The OSIPI pipeline inventory paper states: "*the inventory contains pipelines for human brain data only, as nobody registered ASL pipelines that would be primarily non-brain or non-human.*"
- OSIPI's own **2023–2025 roadmap** says it in writing: "*We do not have any pipelines able to process human clinical non-brain data. While there are many applications for kidneys, liver, but also prostate, breast, and muscles emerging, there is no large free available pipeline for processing these data, and the data evaluation … has to rely on in-house scripts.*"

UKAT (UKRIN Kidney Analysis Toolbox) is the flagship renal MRI package and the only tool listed on renalmri.org's software page. Its mapping modules are B0, diffusion, MTR, T1, T2, T2-stimfit, T2* — **no ASL**. Its entire QA subpackage is one file, `qa/snr.py`, containing **two** classes (`Isnr` and `Tsnr` — the latter regresses out linear and quadratic temporal drift via a GLM, citing Hutton 2011), and its only hard-coded numeric is `tsnr_map[tsnr_map > 1000] = 0`, a divide-by-near-zero clamp, not a quality verdict. A `perfusion.py` exists **only on the `dev` branch** (4,781 bytes, never released — newest tag is v0.7.3, 2024-10-30); it computes `mean_label - mean_control` with no M0, no kinetic model, no units, and opens with `warnings.warn('Perfusion is still a work in progress and likely to be re-written. Use with caution.')`.

> 🎯 **Consequence for osipy-qc:** a kidney QC tool **cannot consume a standard pipeline's derivatives**, because there is no standard pipeline. It must take near-raw NIfTI (4D control/label series + M0 + optionally a perfusion map) plus an **externally supplied** cortex mask.

7.8 OSIPI has no renal anything — but your mentors own the gap

- The peer-reviewed **OSIPI ASL Lexicon** (Suzuki et al. 2024, *MRM* 91(5):1743–1760 — **Sudipto Dolui is author #4**) is brain-scoped: "*(Pseudo-) Continuous ASL, Pulsed ASL, Velocity-selective ASL and Multi-timepoint ASL **for brain perfusion imaging***" — with an explicit invitation to extend: "*the content of the lexicon is **not intended to be limited** to these techniques, and this paper provides the foundation for a growing online inventory that will be extended by the community.*"

Suzuki 2024 OSIPI Lexicon — scope statement

- The **living** (Google Doc) version has a "*Section 7: Outside of Brain*" with kidney cortex/medulla λ and T1 rows and the RBF definition — the only OSIPI-branded renal ASL content that exists.
- **The bridge:** the Lexicon lists "*Technical recommendations for renal ASL from an international group of experts working under the framework of the PARENCHIMA project*" as **one of its four foundational standardisation sources**, alongside ASL-BIDS and the White Paper. OSIPI already recognises the renal consensus.
- **TF2.2** (ASL code library) is led by **María Mora Álvarez**, and states: "*We continuously welcome new ASL code contributions! At the moment we have a special interest in extending our repository with code for preclinical and non-brain code snippets.*"
- **TF1.1** (pipeline inventory) is co-led by **Sudipto Dolui**, and "*is currently expanding the ASL pipeline inventory by including … pipelines specifically targeted to preclinical data and body ASL other than the brain.*"

> ✅ **Say this to the mentors:** "*OSIPI has documented this gap in its own roadmap, and both task forces that own it are led by you two. A renal QC module is not me inventing a niche — it is filling one OSIPI has already written down.*"

7.9 A summary of what can be tagged what

RENAL QUANTITY	BEST AVAILABLE TIER
Acquisition parameter bands (PLD, TI, TI1, τ, TR, slice/voxel size, R≤2, orientation, ≥20 pairs)	✅ `published` — Nery 2020, with per-statement agreement %
M0 mandatory / no BS / TR > 5 s / correction formula	✅ `published`
λ, $T1_{blood}$, α, ×0.93 per BS pulse	✅ `published` (λ flagged as brain-borrowed; blood T1 as `published-via-consensus`)
Report cortex, L/R separately; medulla not reliable	✅ `published`
Registration required; outlier rejection required	✅ `published` *(as requirements — no method, no number)*
Table 4 minimum reporting set	✅ `published`
The five ±SD outlier rules; 750 ms ATT default; 0–500 voxel clip; 8 mm navigator window	⚠️ `implementation`
tSNR / spatial SNR / PWS / BS-efficiency floors	❌ `uncalibrated`
Spatial CoV, negative-voxel fraction, motion (mm or FD analogue)	❌ `uncalibrated` — no renal precedent at all
Cortex:medulla ratio trip point	❌ `uncalibrated` — and it must flag the *segmentation*, not the map
Any absolute cortical RBF band	❌ `uncalibrated` — implausibility guard only
Any medullary threshold	⛔ **do not ship** — R10.2 + ICC 0.08
Any paediatric renal threshold	❌ `uncalibrated` — ~50 children total, zero healthy
Any renal QEI analogue	❌ `uncalibrated` and novel

PART 8 — What this all means for osipy-qc kidney

8.1 The eight design decisions this evidence forces

1. **Branch on labelling scheme first.** FAIR and pCASL are both consensus-endorsed (R3.1) and differ by ~80% in cortical RBF. Every band, every geometry check, every α is scheme-conditional.
2. **Cortex is the ROI; left and right are separate outputs.** R10.1, 100% agreement. Report mean + SD per kidney, plus median when the distribution is skewed. **Medulla is INFO/N-A.**
3. **Expect an externally supplied mask.** Manual ROI is the consensus default (R9.10) and no tool produces cortex/medulla masks. QC the *supplied* mask; gate any cortex/medulla check on its presence.
4. **Stream A is where the value is.** The consensus gives ~20 genuinely `published`, agreement-graded acquisition and protocol rules. Stream B gives none. A renal v1.0 should be overwhelmingly protocol-conformance checks, with perfusion-level checks confined to wide implausibility guards.
5. **Map agreement percentage onto verdict severity.** 100%-agreement, 0%-abstention statements (R8.1, R8.2, R10.1) can drive a FAIL. 75–80% statements (R6.4 SE-EPI, R7.4 free breathing, R7.2 BS) should WARN at most. R5.10 (40% abstention, internally contradictory) is advisory only.
6. **Motion is per-kidney, anisotropic, and respiratory.** Two rigid bodies, not one. Weight the cranio-caudal axis. Do not port an isotropic framewise-displacement scalar.
7. **Verdicts are per-slice, position-weighted.** Coverage loss is normal (48.3% retained all five slices in healthy young adults) and edge slices are systematically worse.
8. **Port Module 6 (M0) nearly unchanged; do not port Module 1 (QEI).** The M0 rule is identical including the 5 s threshold. QEI has no tissue priors to stand on.

8.2 What ports, what doesn't

BRAIN MODULE	PORTS TO KIDNEY?	NOTE
1 — QEI engine	❌ **No**	No tissue probability maps, no rated renal dataset
2 — Noise & distribution	⚠️ Partly	Negative-voxel fraction ports onto a cortex mask; sCoV has no renal precedent and is confounded by the cortico-medullary gradient
3 — CBF level & contrast	⚠️ Reframe	Level → wide implausibility guard only. GM/WM ratio → cortex:medulla as a **segmentation-integrity** flag
4 — Co-registration	✅ Mostly	Dice/Pearson/overlap are organ-agnostic and take arbitrary binary masks. Needs a kidney mask; must run **per kidney**
5 — Schema & data-type	⚠️ Rebuild	BIDS has no renal contract. Build against **Nery Table 4** instead
6 — M0 calibration	✅ **Yes**	Same rule, same 5 s, same formula. Swap $T1_{tissue}$, evaluate per compartment

BRAIN MODULE	PORTS TO KIDNEY?	NOTE
7 — Motion	⚠️ Rebuild	Respiratory, per-kidney, anisotropic. The five ±SD outlier rules are the renal analogue
8 — Acquisition metadata	✅ **Yes, richer than brain**	~20 agreement-graded consensus values to check against

8.3 Open questions to put to Maria's renal specialist

1. **Which cortex mask should the toolbox expect?** Manual (consensus default), Renal Segmentor whole-kidney + erosion, or a T1-map-derived cortex? This determines whether any cortex/cortex-ratio check is even runnable.
2. **Is a cortex:medulla segmentation-integrity flag welcome, given R10.2?** It uses medulla as a denominator to sanity-check the *segmentation*, not to report medullary perfusion.
3. **What is a realistic renal FAIL rate to design against?** Radovic's 14.3% (paediatric transplant) and Olsen's 51.7% of scans losing ≥1 slice are the only measured numbers I have.
4. **Should the ≥20-pair check be readout-gated?** The rule is scoped in the body text to 2D readouts, and the paediatric 3D-GRASE studies used 6 and 25.
5. **What kidney T1 should the M0 correction assume**, and should it be per-compartment?
6. **Would the task force accept a renal extension of the OSIPI Lexicon's "Section 7"?**

8.4 Two things I could not read, and should

- **Taso M, Aramendía-Vidaurreta V, Englund EK, Francis S, Franklin SL, Madhuranthakam AJ, et al. "Update on state-of-the-art for arterial spin labeling (ASL) human perfusion imaging outside of the brain."** ***Magn Reson Med*** **2023;89(5):1754–1776. DOI 10.1002/mrm.29609.** This is an **ISMRM Perfusion Study Group** review whose organ-by-organ section **starts with kidneys** and which states "*Summaries and recommendations of acquisition parameters (when appropriate) are provided for each organ.*" Its senior author is **María Fernández-Seara — the same senior author as the Nery consensus.** Unpaywall reports it CC-BY, but Wiley's Cloudflare blocks scripted access and there is no PMC copy. **Download it in a browser.** Until then, treat renal acquisition guidance in this document as possibly incomplete.
- **Mora Álvarez MG, Madhuranthakam AJ, Udayakumar D. "Quantitative non-contrast perfusion MRI in the body using arterial spin labeling."** ***MAGMA*** **2024. DOI 10.1007/s10334-024-01188-1.** Paywalled — **and written by your own mentor.** Ask her directly.

📎 Appendix A — Citation errors I found and corrected

Caught while verifying that each downloaded PDF was the paper it claimed to be. Full table in `papers/README.md`; the ones most likely to embarrass you in a meeting:

WRONG	RIGHT
"Cox EF et al." for the *Kidney Int Rep* CKD RBF paper	**Li L-P, Tan H, Thacker JM, Li W, Zhou Y, Kohn O, Sprague SM, Prasad PV** — a different group entirely. The numbers were right; the attribution was not
"Hueper K / Nery F" for the 1.5T reproducibility paper	**Hammon M, Janka R, Siegl C, Seuss H, Grosso R, Martirosian P, Schmieder RE, Uder M, Kistner I**, *Medicine* 2016;95(11):e3083
"Bones / Nery" for the 2D readout comparison	**Buchanan CE, Cox EF, Francis ST**, *Diagnostics* 2018;8(3):43
"Cox 2019" for the furosemide cortex/medulla study	**Haddock B, Larsson HBW, Francis S, Andersen UB**, *Acta Physiol* 2019;227:e13292
"Cox 2018 Eur Radiol" / "Kim 2018" for the 3D-GRASE multi-TI study	**Shirvani S, Tokarczuk P, Statton B, … O'Regan DP**, *Eur Radiol* 2019;29:232–240
"Franklin SL first author" for MRM 2020;84:1919–1932	**Bones IK** is first author
Nery 2020 has "166 consensus statements"	**59** (p.6: "*The 59 final consensus statements are listed in Table 2*")
Nery 2020 panel of "24 authors"	**26 authors, 23-member expert panel**
Harteveld 2020 readout = SE-EPI	**single-shot gradient-echo EPI 2D multislice** — and the consensus recommends *spin*-echo, so the flagship FAIR-vs-pCASL comparison used a non-recommended readout
Odudu 2018 Table S1 "404s"	Real: **969 rows × 23 columns**, retrievable via the Europe PMC supplementaryFiles API, saved at `papers/supplementary/odudu2018_supp_table_s1_53_studies.xlsx`

Sources

The two anchor documents

- **Nery F, Buchanan CE, Harteveld AA, Odudu A, Bane O, Cox EF, Derlin K, Gach HM, Golay X, Gutberlet M, Laustsen C, Ljimani A, Madhuranthakam AJ, Pedrosa I, Prasad PV, Robson PM, Sharma K, Sourbron S, Taso M, Thomas DL, Wang DJJ, Zhang JL, Alsop DC, Fain SB, Francis ST, Fernández-Seara MA.** "*Consensus-based technical recommendations for clinical translation of renal ASL MRI.*" **MAGMA 2020;33(1):141–161.** DOI 10.1007/s10334-019-00800-z · CC-BY · `papers/nery2020_renal_asl_consensus.pdf`
- **Odudu A, Nery F, Harteveld AA, Evans RG, Pendse D, Buchanan CE, Francis ST, Fernández-Seara MA.** "*Arterial spin labelling MRI to measure renal perfusion: a systematic review and statement paper.*" **Nephrol Dial Transplant 2018;33(suppl_2):ii15–ii21.** DOI 10.1093/ndt/gfy180 · `papers/odudu2018_renal_asl_systematic_review.pdf`

Consensus family

- Mendichovszky I, Pullens P, Dekkers I, et al. *MAGMA* 2020;33(1):131–140 — the PARENCHIMA Delphi process paper
- Dekkers IA, de Boer A, Sharma K, et al. *MAGMA* 2020;33:163–176 — renal T1/T2 consensus, the only one addressing cortex-vs-medulla ROI methodology directly

Healthy reference and reproducibility

- Garcia-Ruiz L, Echeverria-Chasco R, Aramendia-Vidaurreta V, et al. *JMRI* 2025;62(4):1180–1195 — field strength, sex, age; the QC-metric table
- Harteveld AA, de Boer A, Franklin SL, Leiner T, van Stralen M, Bos C. *MAGMA* 2020;33(1):81–94 — FAIR vs pCASL, same subjects
- Buchanan CE, Cox EF, Francis ST. *Diagnostics* 2018;8(3):43 — readout comparison
- Gillis KA, McComb C, Foster JE, et al. *BMC Nephrol* 2014;15:23 — 3T inter-study reproducibility
- Hammon M, Janka R, Siegl C, et al. *Medicine* 2016;95(11):e3083 — 1.5T reproducibility (⚠️ medulla contaminated)
- Haddock B, Larsson HBW, Francis S, Andersen UB. *Acta Physiol* 2019;227:e13292 — the credible cortex:medulla split
- Olsen NE, Mariager CO, Arildsen MM, et al. *Magn Reson Med* 2025 — ASL vs $[^{15}O]H_2O$ PET
- Cox EF, Buchanan CE, Bradley CR, et al. *Front Physiol* 2017;8:696 — n=127 healthy cohort
- Shirvani S, Tokarczuk P, Statton B, et al. *Eur Radiol* 2019;29:232–240 — multi-TI vs single-TI
- de Boer A, Harteveld AA, Stemkens B, et al. *JMRI* 2021;53:859–873 — multiparametric test-retest
- Wolf M, de Boer A, Sharma K, et al. *Nephrol Dial Transplant* 2018 — renal T1/T2 systematic review

Motion, labelling and artefact

- Bones IK, Harteveld AA, Franklin SL, et al. *Magn Reson Med* 2019;82(1):276–288 — free-breathing BGS, the 54% registration finding
- Bones IK, Franklin SL, Harteveld AA, et al. *Magn Reson Med* 2020;84:1919–1932 — spurious VS labelling, the bellows finding

- Bones IK, Franklin SL, Harteveld AA, et al. *Magn Reson Med* 2021;86:131–142 — VSASL label dynamics, FAIR ATT
- Bones IK, Bos C, Moonen C, Hendrikse J, van Stralen M. *Magn Reson Med* 2022;87:800–809 — the 3-U-net cascade
- Franklin SL, Bones IK, Harteveld AA, et al. *Magn Reson Med* 2021;85:2580–2594 — multi-organ tSNR comparison
- Yamashita H, Yamashita M, Futaguchi M, et al. *SpringerPlus* 2014;3:131 — 4D-CT kidney motion (centre of gravity)
- Siva S, Pham D, Gill S, et al. *Radiat Oncol* 2013;8:248 — 4D-CT kidney motion (apex)
- Song H, Ruan D, Liu W, et al. *Med Phys* 2017;44(3):962–973 — respiratory motion prediction
- Brumer I, Bauer DF, Schad LR, Zöllner FG. *Diagnostics* 2022;12(8):1854 — synthetic renal ASL phantom (**ground truth cortex 250 / medulla 50 healthy; 100 / 20 abnormal** — do not confuse with the *recovered* 210/49)
- Zhao W-T, Herrmann K-H, Wei W, et al. *MAGMA* 2025;38:503–517 — the only self-described ASL "quality assurance protocol" outside brain (rat liver, 9.4T)

Disease, transplant and paediatric

- Li L-P, Tan H, Thacker JM, et al. *Kidney Int Rep* 2017 · Zhang X, Ye C, Lu F, et al. *Renal Failure* 2024;46(2):2428337 · Shi R, Wang H, Xu H, et al. *Insights Imaging* 2026;17:35 · Zhao L, Xu FB, Liu JY, et al. *Renal Failure* 2025;47(1):2444403 · Fang W, Song Q, Cao F, et al. *QIMS* 2026;16(4):299
- Echeverria-Chasco R, Martin-Moreno PL, Garcia-Fernandez N, et al. *NMR Biomed* 2023;36(2):e4832 · Que C, Zhang H, Ma J, et al. *QIMS* 2026;16(2):165 · Ma J, Cao C, Que C, et al. *QIMS* 2025;15(8):6882–6896 ⚠️ · Peng J, et al. *Transl Androl Urol* 2023;12(4):612–621 · Jiang B, Li J, Wan J, et al. *QIMS* 2024;14(3):2415–2425 · Chhabra J, et al. *Cureus* 2022
- **Nery F, De Vita E, Clark CA, Gordon I, Thomas DL.** *Magn Reson Med* 2019;81:2972–2984 — the only dedicated paediatric native-kidney renal ASL study
- Harteveld AA, Littooij AS, van Noesel MM, et al. *MAGMA* 2022;35:235–246 — paediatric abdominal pCASL
- Radovic T, Spasojevic B, et al. *Sci Rep* 2022;12:828 · Radovic T, Spasojevic B, Cvetkovic M, et al. *Sci Rep* 2025
- De Mul A, et al. *Pediatr Nephrol* 2025;40(5):1539–1548 — "*experimental and not validated*"

Standards, tooling and preclinical

- Clement P, Castellaro M, Okell TW, et al. *Sci Data* 2022;9:543 — ASL-BIDS ("*validated in ASL images of the brain only*")
- Suzuki Y, Clement P, Dai W, **Dolui S**, Fernández-Seara MA, et al. *Magn Reson Med* 2024;91(5):1743–1760 — OSIPI ASL Lexicon
- Fan H, Mutsaerts HJMM, Anazodo U, … **Dolui S**, Petr J. *Magn Reson Med* 2023 — OSIPI ASL pipeline inventory
- Nery F, Gordon I, Thomas DL. *Diagnostics* 2018;8(1):2 — renal ASL challenges and opportunities review
- Chuang K-H, Kober F, Ku M-C. "*Quantitative Analysis of Renal Perfusion by ASL*" and Chuang K-H, Meier M, Fernández-Seara MA, Kober F, Ku M-C. "*Renal Blood Flow Using ASL MRI: Experimental Protocol and Principles*" — Ch. 39 and Ch. 26 in **Preclinical MRI of the Kidney** (Methods Mol Biol 2216, 2021), CC-BY
- Hillaert A, Sanmiguel Serpa LC, Xu Y, et al. *Animals* 2024;14(12):1810 — λ measured in dog kidney

- **UKAT** — github.com/UKRIN-MAPS/ukat (not cloned; depends on scikit-learn, which osipy's no-scipy rule would forbid)
- Gullaksen S, et al. *Diabetologia* 2023;66:813–825 — the true origin of the 0–500 mL/100 g/min voxel rejection rule

Could not obtain — 22 papers listed with DOIs in `papers/README.md`, including Robson 2009 (the bellows-sorting algorithm and COVF metric), Tan 2014 (the 8 mm navigator window), both Artz 2011 papers, Cutajar 2012, Kim 2017, Ahn 2020, de Bazelaire 2004, Zhang 2013 (the primary source for both blood-T1 constants), and **Taso 2023** — which is CC-BY and should simply be downloaded in a browser.

Every screenshot above lives in `screenshots/` at 130 dpi, rendered with `pdftoppm -r 130 -png -f N -l N`. Every PDF lives in `papers/`. Diagrams are inline Mermaid.